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EACTS Expert Consensus Statement on the Ross Procedure in Adult Patients

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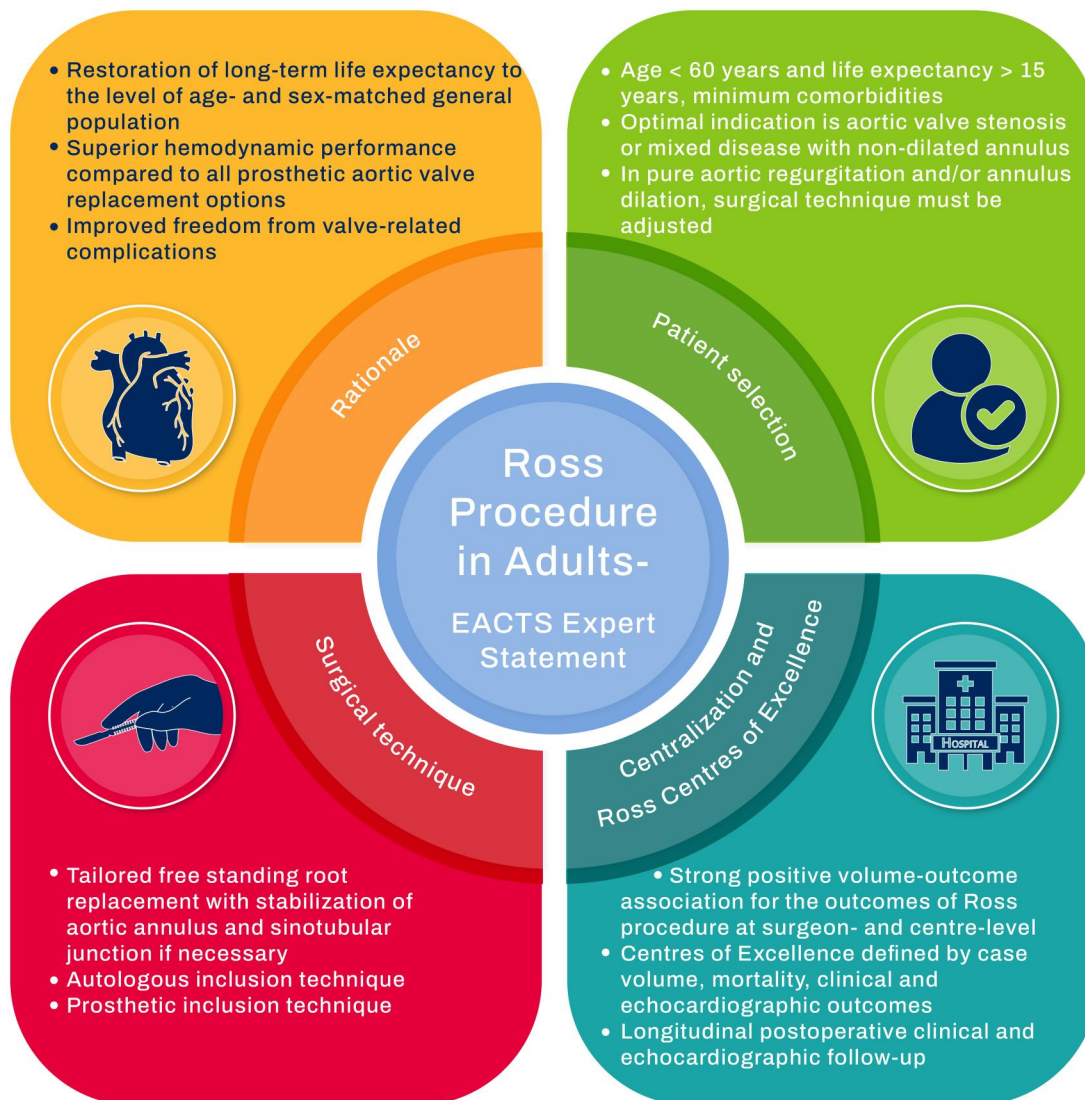
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Graphical abstract



Central Illustration. The Ross Procedure in Adults: Rationale, Patient Selection, Technique, and Centralization of the Procedure.

Abstract

The Ross procedure, which involves replacing a diseased aortic valve with the patient's own pulmonary valve (pulmonary autograft), has gained renewed attention in adult patients due to mounting evidence of excellent long-term outcomes. With increasing data demonstrating restoration of life expectancy and improved valve-related outcomes, it is now considered a viable first-line option for selected young and middle-aged adults with non-repairable aortic valve disease. Nonetheless, its adoption remains inconsistent across institutions, partly due to concerns about surgical complexity, long-term durability, and the need for structured follow-up. This expert consensus document, commissioned by the European Association for Cardio-Thoracic Surgery, synthesizes the current evidence and offers clinical statements for using the Ross procedure in adults. Developed through a systematic review and informed by the collective experience of a multidisciplinary panel of internationally recognized experts, the document addresses key topics including patient selection, technical refinements, and perioperative management. Emphasis is also placed on the need for procedural standardization, high-volume surgical expertise, and longitudinal imaging-based follow-up. Moreover, comparative outcome data and subgroup considerations are critically appraised, and areas for future investigation are outlined. This clinical practice document is intended to assist clinicians and Heart Teams in making informed treatment decisions, support harmonized implementation across centres, provide technical considerations to optimize procedural success, and substantially improve outcomes for patients considered for this complex but promising procedure.

Keywords: Ross procedure; aortic valve replacement; pulmonary autograft; expert consensus statement; clinical practice document; adult cardiac surgery.

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ABBREVIATIONS AND ACRONYMS

AR	Aortic regurgitation
AV	Aortic valve
AVR	Aortic valve replacement
bAVR	Bioprosthetic aortic valve replacement
CT	Computed tomography
EACTS	European Association for Cardio-Thoracic Surgery
GRR	German Ross Registry
LAD	Left anterior descending artery
LV	Left ventricle
LVOT	Left ventricular outflow tract
mAVR	Mechanical aortic valve replacement
MFS	Marfan syndrome
PA	Pulmonary autograft
PEARS	Personalised External Aortic Root Support
PPM	Patient-prosthesis mismatch
PV	Pulmonary valve
RCT	Randomized controlled trial
RR	Root replacement
RR+R	Reinforced root replacement
RVOT	Right ventricular outflow tract
SC	Subcoronary (technique)
SDM	Shared decision making
STJ	Sinotubular junction
TOE	Transoesophageal echocardiography

1. INTRODUCTION

The choice of an optimal valve substitute for a non-elderly patient with aortic valve (AV) disease remains difficult. European and American clinical practice guidelines recommend mechanical aortic valve replacement (mAVR) as a preferred option in patients below 50 to 60 years of age, and selectively in older individuals.^{1,2} Despite effective treatment of valve pathology, patients receiving mAVR are exposed to a clinically relevant long-term risk of thromboembolism and major haemorrhage, concerns with quality of life, and, ultimately, a survival lower than the general age- and sex-matched general population.³⁻⁷ Aortic valve replacement with bioprosthetic valves (bAVR) obviates the need for lifelong anticoagulation and has been increasingly used in this patient cohort. Nevertheless, despite a lower risk of bleeding and thromboembolic events compared to mAVR, bAVR is associated with limited durability, especially in younger individuals, and excess long-term mortality compared to the age- and sex-matched general population.^{5,6,8-11}

Over the last 15 years, there has been a renewed interest in the Ross procedure as an alternative to conventional AV substitutes in non-elderly patients. A growing body of evidence has consistently shown that, unlike prosthetic valves, the Ross procedure is the only AV replacement option that results in restored long-term patient survival compared to that of the general population.^{6,12-15} This outcome is in large part a result of replacing the diseased AV with the patient's own living pulmonary valve (PV), which translates into excellent haemodynamics, low

thrombogenicity, and resistance to infection.^{16,17} However, concerns remain about the technical complexity of the operation, which may lead to suboptimal early and late outcomes when performed in low-volume centres or by inexperienced surgeons.^{18,19} Additionally, the potential long-term valve failure in both aortic and pulmonary positions is a matter of concern.^{13,14,20,21} As a result, various technical modifications and clinical recommendations have emerged to reduce those risks and improve long-term patient outcomes.²²⁻²⁵

Current European guidelines for the management of valvular heart disease provide limited guidance about the role of the Ross procedure in adults, limiting its use to selected individuals with AV stenosis in experienced centres as an alternative to valve replacement.² The growing interest and promising long-term outcomes of the Ross procedure reported worldwide, however, demand a more detailed and up-to-date statement. This consensus document aims to provide the European Association for Cardio-Thoracic Surgery (EACTS) expert consensus on the role of the Ross procedure in adult patients. Specifically, our objectives are to critically analyse the contemporary evidence on the use of the Ross procedure in adults, to define the suitable candidate population, highlight the crucial technical steps, describe the specifics of long-term patient management and possible reinterventions, and to provide a platform for further clinical decision making and future trends in this area.

2. METHODOLOGY

An expert task force conducted a systematic evaluation of the Ross procedure to assess its role as a treatment option for various forms of AV disease and different patient populations, with the aim of providing evidence-based guidance for health-care providers and researchers. This task force, composed of leading professionals in the field, was appointed by the EACTS Committee for Practice Guidelines, following the procedures outlined in the *EACTS Methodology Manual for Clinical Practice Documents*.²⁶ This document represents the official position of EACTS on this topic.

The composition of the task force prioritized diversity, transparency, and expertise, with members required to disclose any industrial or external affiliations. All participants submitted conflict of interest declarations at the project's initiation and updated them throughout the drafting phase as needed. Once the project scope and content were defined, members without relevant conflicts of interest were assigned to draft specific sections. A literature screening was conducted using the PICOT framework (Population, Intervention, Comparison, Outcome, Time), with searches performed in MEDLINE, Embase, and the Cochrane Library. Only English-language studies focusing on adult cardiac surgery were included. All PICOTs are provided in the [online Supplementary Material](#).

Following the literature screening, consensus statements and supporting explanations were developed in accordance with the *EACTS Methodology Manual for Clinical Practice Documents*. All sections were collaboratively drafted, with key consensus statements shaped through expert discussions during task force meetings. Given the significant variability in economic factors and the lack of cost-effectiveness data, economic analyses were not performed. This document focuses solely on best practices.

To uphold rigour and transparency, a structured anonymous voting process was implemented. Task force members reviewed statements, figures, and algorithms developed in previous meetings, providing both votes of approval or disapproval and additional comments. A minimum of 75% agreement was required for statement

approval, and at least 80% of the members had to participate in the voting process for the results to be considered valid. The final draft underwent internal validation by the writing committee and external peer review by anonymous reviewers selected by the Editor-in-Chief of the *European Journal of Cardio-Thoracic Surgery* (EJCTS) and the EACTS Clinical Practice Guideline Committee, followed by endorsement from the EACTS governing bodies.

3. RATIONALE FOR THE ROSS PROCEDURE

The Ross procedure is the only aortic valve replacement (AVR) operation that is accompanied by the long-term viability of the native aortic valve and root. Microscopic examination of pulmonary autograft (PA) explants years after the Ross procedure shows the presence of a preserved trilaminar structure with endothelial and interstitial cells within the neo-aortic valve cusps.²⁷ In addition to being a living substitute, the PA preserves the dynamism and coupling of the left ventricular outflow tract (LVOT)-aortic root complex.²⁸ These unique biological and haemodynamic characteristics are thought to portend many of the observed clinically relevant benefits of the Ross procedure in adults.

3.1 Survival

In 2005, Yacoub et al. suggested the potential for restored late survival after the Ross procedure in young adults, using microsimulation models.²⁹ Since then, many different cohort studies from various countries and continents have consistently shown the same outcome: restored long-term survival compared to the age- and sex-matched general population (**Table 1**). In the largest cohort study stemming from the German Ross Registry (GRR), including 2,444 patients from 10 European centres, the long-term survival after the Ross procedure was comparable to that of the age- and sex-matched German general population.¹³ Similar findings were illustrated in various cohort studies of adult patients undergoing the Ross procedure.^{6,12,20,30-34} The longest follow-up study published to date by Yacoub's group showed continued survival benefit up to 25 years after the Ross procedure in a cohort of 108 patients undergoing the Ross procedure in the United Kingdom.³⁵ Importantly, this cohort was a part of a randomized controlled trial (RCT) and included a wide array of clinical scenarios: 8% of patients had active endocarditis, 42% had reoperations, and 45% had isolated aortic regurgitation (AR).¹⁵

In contrast, analyses of long-term survival after isolated bAVR or mAVR in young and middle-aged adults systematically show excess mortality late after the operations compared to the age- and sex-matched general population, including cohorts who underwent elective isolated AVR.³⁶⁻³⁹

Although most comparative studies examining the Ross procedure versus alternative AVR options are propensity-matched analyses, 1 RCT compares the Ross procedure to homograft root replacement.¹⁵ In that study of 216 patients (mean age 38 years), the 12-year survival after the Ross procedure was significantly better than that after homograft root replacement. All propensity-matched analyses have demonstrated comparable findings. In a study from Australia comparing the Ross procedure to mAVR, there was a 10% absolute survival benefit at 20 years in the matched cohort.³⁰ Similar findings were obtained when the Ross procedure was compared to bAVR in a Canadian adult population (mean age 44 years).³² At 20 years, mortality after bAVR was 25% versus 9.6% after the Ross procedure.

Table 1. Studies Reporting Long-term Survival Following the Ross Procedure in Adults

Author	Year	Sample size	Patient population	Surgical technique, %	In-hospital deaths, %	Long-term survival, %
Aboud ¹³	2021	2444	German Ross Registry	SC 36.5, RR 19.6, RR+R 43.9	1	95.4 at 10 y, 79.4 at 20 y, 75.8 at 25 y
Oeser ⁴⁰	2013	224	Single-centre	RR 69.2, reinforcement/ various RI forms 27.2	1.8	87.3 at 20 y
Buratto ³⁰	2016	392	Single-centre	Autologous RI	0.3	98 at 10 y, 95 at 20 y
Da Costa ⁴¹	2014	414	Single-centre	RR 86, prosthetic RI 14	2.7	89.3 at 15 y
David ¹²	2019	212	Single-centre	SC 49, RR 51	0.5	89.2 at 20 y
El-Hamamsy ⁶	2022	434	California and New York state registry	N/A	0.23	93.2 at 15 y
Gofus ¹²	2022	291	Czech Republic National Registry	RR+R	0	93.6 at 10 y
Martin ³¹	2017	310	Single-centre	RR 83.6	1.3	94.1 at 10 y, 83.6 at 20 y
Mastrobuoni ⁴³	2016	306	Single-centre	SC 2.3, RR 54.9, prosthetic RI 42.8	2.3	87 at 20 y
Mazine ³²	2022	108	Single-centre	N/A	0	96.9 at 10y, 90.4 at 20 y
Notenboom ³⁵	2024	108	Single-centre (RCT)	RR	0.9	83 at 25 y
Romeo ²⁰	2021	1431	5 high-volume centres	SC 41.1, RR+R 34, RI 24.9	0.7	95.1 at 10 y, 88.5 at 15 y
Ryan ³³	2021	225	Single-centre	RR+R	2.2	94.2 at 10 y, 81.3 at 20 y
Sharabiani ⁴⁴	2016	718 (of those 494 children)	United Kingdom National Registry	N/A	1.1	97.3 at 12 y
Tsaroev ⁴⁵	2022	215	Single-centre	RR 74, 26 autologous or prosthetic RI	0.9	94.7 at 12 y

Abbreviations: GRR, German Ross Registry; N/A, not available; RCT, randomized controlled trial; RI, root inclusion; RR, root replacement; RR+R, reinforced root replacement; SC, subcoronary; y, year.

Additionally, in a study examining long-term outcomes after the Ross procedure, mAVR or bAVR using California and New York State data, survival after the Ross procedure was comparable to that of the age-, sex- and race-matched general population at 15 years, and significantly better than that after mAVR or bAVR.⁶

In summary, it is now well established that the Ross procedure is associated with life expectancy comparable to that of the age- and sex-matched general population. Additionally, survival after the Ross procedure is superior to that after conventional prosthetic AVR, likely as a result of better haemodynamics, a lower rate of valve-related complications (endocarditis, thromboembolism, and bleeding), a lower rate of pacemaker implants, and excellent durability when performed by experienced surgeons.

3.2 Haemodynamics

Patient-prosthesis mismatch (PPM) has a disproportionately worse impact in younger patients compared to their elderly counterparts.⁴⁶ This outcome is likely a result of their higher physiological requirements. The presence of PPM imposes an afterload burden on the left ventricle (LV), which has an additional negative impact on valve durability as well as patient survival in the long term.^{47,48}

One of the unique features of the Ross procedure is the excellent haemodynamic performance of the PA, including in patients with a small aortic annulus. In a study by Pibarot et al.,⁴⁹ mean transvalvular gradients across the PA were similar to those of healthy individuals at rest. Importantly, when patients were subjected to maximal exercise thresholds, mean aortic gradients remained unchanged (<10 mmHg) and comparable to those of healthy controls. The Ross procedure offers a significantly larger effective orifice area for the same aortic annulus size compared to the corresponding bAVR alternative, thus further contributing to excellent haemodynamics and minimizing the risk of PPM.³² Indeed, the risk of severe PPM in a PA is nearly non-existent.

In another study examining spatiotemporal flow patterns late after the Ross procedure using 4-dimensional magnetic resonance imaging, there were no significant differences compared to healthy controls.⁵⁰ Lastly, unlike bioprostheses, the PA does not degenerate through calcification and the development of stenosis.⁵¹ As a result, the mean aortic gradient remains unchanged over time, which further contributes to optimal long-term LV dynamics.²⁰ In a meta-analysis by Um et al., the Ross procedure was associated with significantly lower transaortic gradients than any other aortic valve substitute at discharge; in the long-term postoperatively, the same benefit was observed in the aortic orifice area.¹⁶ There is no need for aortic annulus enlargement in adult patients undergoing the Ross procedure, even in the case of an enlarged pulmonary annulus compared to an aortic annulus.

On the other hand, a pulmonary homograft remains at some risk of developing stenosis, particularly in the first years postoperatively.^{20,52–54} It follows a biphasic pattern with an incidence of 3.5% in the first year and then falls to 1% per year in the long term. Early failures could be related to technical issues, or more likely, to the immune response that can affect the proximal and distal suture line, cusps, and the body of the homograft, thus leading to pulmonary stenosis. Echocardiographic studies have shown that, unlike the PA in the aortic position, a homograft in the pulmonary position develops some degree of stenosis compared to healthy controls at rest and during exercise.⁴⁹

3.3 Valve-related complications

As a living valve substitute, the rates of valve-related complications after the Ross procedure are low. No anticoagulation is necessary after the Ross procedure, thus limiting the incidence of bleeding events. Additionally, the incidence of thromboembolic events is very low. In a study by El-Hamamsy et al., the cumulative incidence of major bleeding or stroke was comparable after the Ross procedure (1.9% and 2.1% at 15 years postoperatively) and bAVR (3.3% and 3.3% at 15

years, respectively), and significantly lower than after mAVR (5.2% and 4.8% at 15 years, respectively).⁶ Similarly, an extensive network meta-analysis by Yokoyama et al. that included 4812 patients,¹⁰ concluded that the Ross procedure provides superior freedom from stroke compared to bAVR and mAVR and superior freedom from bleeding events compared to mAVR.

The living autologous tissue in the aortic position provides excellent resistance against valve infection, and autograft endocarditis is very rare. In contrast, there is a small but constant risk of pulmonary homograft infection in the long-term postoperative period. In the RCT of the Ross procedure versus homograft aortic root replacement, the Ross procedure resulted in significantly better freedom from endocarditis at 12 years (97% vs 82%, respectively).¹⁵ No case of autograft endocarditis and 2 cases of late pulmonary homograft endocarditis were reported in this study. Similar results were

However, recent reports from dedicated centres worldwide dispel many of these concerns. In a multicentre study from 6 high-volume centres by Romeo et al., freedom from reintervention on either autograft or homograft was 90.8% at 15 years postoperatively.²⁰ A GRR analysis by Aboud et al. reported 84.7% 15-year freedom from any Ross-related reintervention.¹³ Other studies have shown a similar trend of about 1%/patient-year risk of reintervention, which is significantly lower than that observed after bAVR.^{6,32,34,58} It is notable that in the first decade of follow-up, the reintervention rate after bAVR is similar to that after the Ross procedure. However, in the second decade of follow-up, reinterventions after bAVR show a steep increase and by far exceed those after the Ross procedure. Furthermore, although reintervention rates after mAVR are generally lower, the difference at 15 years is <10% in most studies of non-elderly adults.^{6,12,34,37,59}

Consensus Statement Table 1. The Rationale for the Ross Procedure in Adults
• The Ross procedure is associated with restoration of long-term life expectancy compared to that of the age- and sex-matched general population. • The Ross procedure provides superior haemodynamic performance compared to all prosthetic AV replacement options. • The Ross procedure is associated with improved freedom from valve-related complications, including endocarditis and PPM, improved freedom from reoperation compared to bAVR, and improved freedom from thromboembolism and haemorrhage compared to mAVR.

Abbreviations: AV: aortic valve; bAVR: bioprosthetic aortic valve replacement; mAVR: mechanical aortic valve replacement; PPM: patient-prosthesis mismatch.

observed in a nationwide analysis by Gofus et al.:¹² no autograft and 2 homograft endocarditis cases were reported, both in active drug abusers. Pulmonary homograft remains a weak point with regard to infection, particularly in active intravenous drug abusers, as demonstrated in the study of Chauvette et al.⁵⁵ Nevertheless, overall rates of endocarditis after the Ross procedure are significantly lower than those after bAVR and are comparable to those after mAVR, as shown in several studies.^{6,10,56} In the absence of Ross-specific evidence, general recommendations for antibiotic prophylaxis are applied in line with current guidelines for other valve substitutes.⁵⁷

There is an inconsistency regarding the rate of postoperative pacemaker implants after the Ross procedure. In a study by Mazine et al., the incidence of pacemaker implants was significantly lower after the Ross procedure compared to that after bAVR in young and middle-aged adults.³² According to Yokoyama, a permanent pacemaker was implanted at a similar rate to those implanted after the Ross procedure and bAVR, but less frequently than those implanted after an mAVR.¹⁰ Other studies did not observe any differences between Ross and mAVR with regard to pacemaker implant rates.³⁴ The incidence is more variable in patients undergoing the Ross procedure as a redo intervention after a previous prosthetic AVR, because removal of the prosthesis can result in damage to the conduction tissue. In primary cases, however, the absence of a rigid sewing ring combined with an adequate surgical technique results in a rate of pacemaker implants typically below 1%.³² The pacemaker implant rate remains low in the early postoperative period and does not increase over time.

3.4 Ross-related reinterventions

The Ross procedure was once dubbed a “Trojan horse” due to a worrisomely high hazard of reoperation in a single high-volume institution.¹⁸ The thought of “transforming a single-valve into double-valve disease” led to a steep decline in its use by the early 2000s.¹⁹

4. PATIENT SELECTION

The success of the Ross procedure is dependent on careful patient selection, which involves a thorough evaluation of patient preferences as well as of anatomical and clinical factors. Not all individuals with AV disease are suitable candidates.¹⁷ Careful assessment of the structure and function of the AVs and PVs is mandatory. Key factors such as valve morphology, size, and the condition of the aortic root play an important role in determining the safety and effectiveness of the Ross procedure as a surgical strategy. This chapter explores the criteria and considerations that guide the selection of patients for the Ross operation (Figure 1).

4.1 Shared-patient decision-making concept

Given the complexity of choosing the best AV substitute in a non-elderly patient, there is a growing emphasis on shared decision making (SDM)—a process in which clinicians and patients work together to make decisions and select tests, treatments, and care plans based on clinical evidence that balances risks and expected outcomes with patient preferences and values.⁶⁰ Interestingly, almost 1 out of 4 non-elderly patients in need of AV replacement were not involved in the choice of type of valve substitute, as illustrated by a questionnaire carried out in centres worldwide.⁶¹

An integral part of SDM is patient information and education. These items can be augmented through the use of apps, websites, and digital technologies, which might help to support the SDM process, as well as engagement with and adherence to the therapy postoperatively. In non-elderly patients with valvular heart disease, use of a decision aid resulted in better informed decisions, less patient anxiety and depression, and better mental well-being.⁶² Careful consideration of individual patient preferences, values, and circumstances is critical.

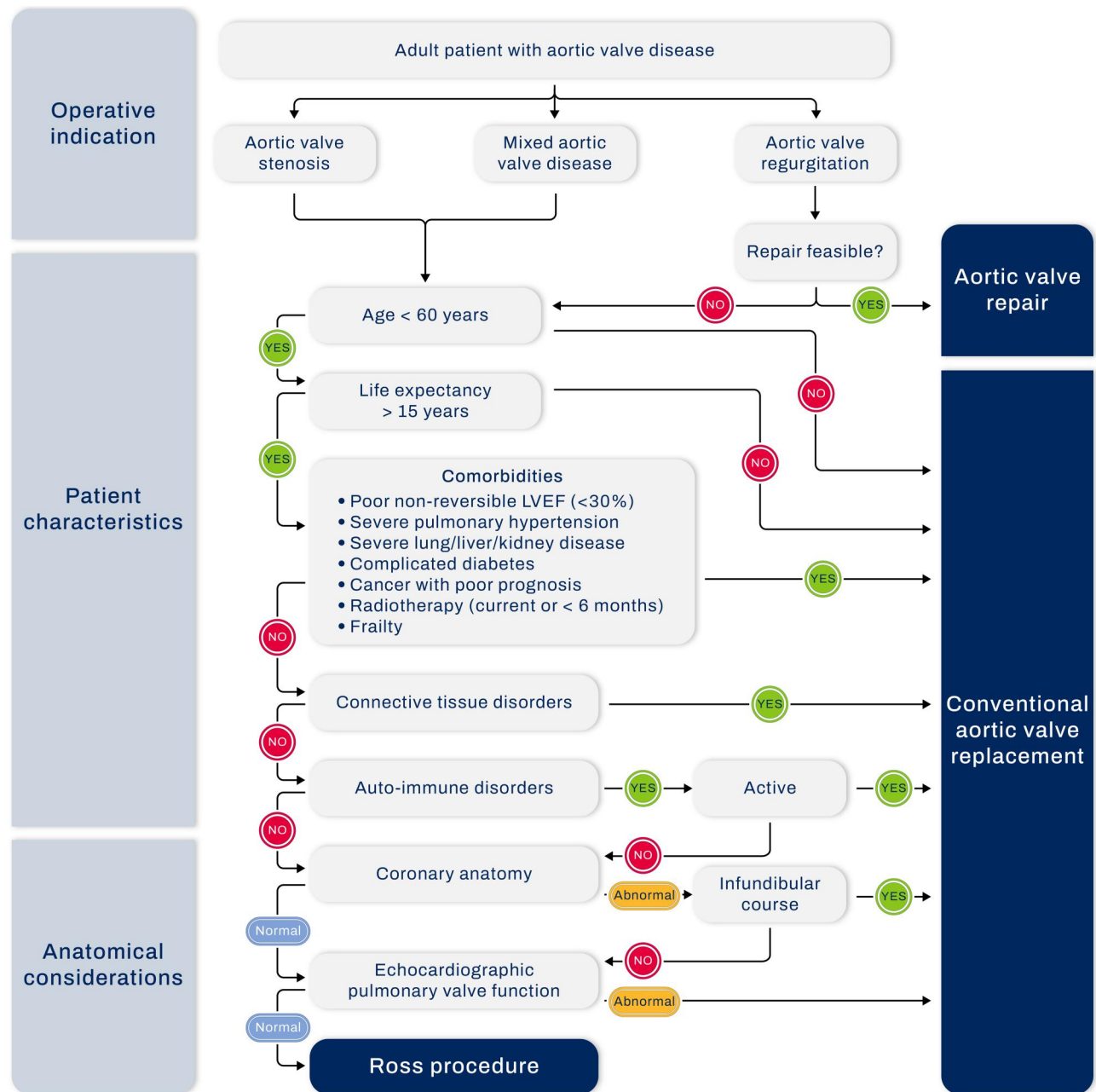


Figure 1. Patient Selection Algorithm. Abbreviation: LVEF, left ventricular ejection fraction

4.2 Patient characteristics and anatomical considerations

4.2.1 Age. The Ross procedure is traditionally offered to young adult patients. The benefits of the procedure manifest over the long term after the index operation. Therefore, this operation should primarily be considered in patients with an anticipated life-expectancy of at least 15 to 20 years. According to the 2020 American College of Cardiology/American Heart Association Guidelines for Management of Patients with Valvular Heart Disease, the Ross procedure is a class IIb recommendation for patients <50 years old with appropriate anatomy.¹ Nevertheless, the average life expectancy of a 50-year-old is more than 30 years in Western countries.^{63,64} Additionally, long-term studies of adults undergoing bAVR or mAVR demonstrate

excess long-term mortality up to the age of 70 at the time of the operation.³⁶

Retrospective studies with appropriately selected patients showed no difference in long-term survival after the Ross procedure in patients below and above 50 years of age up to 10 years postoperatively.⁶⁵ Furthermore, the age range in all of the large Ross cohort studies extends into the early 60s. In all age groups, the Ross procedure restores the expected survival to that of the age- and sex-matched general population.

Mid-term data have shown that the Ross procedure may be performed in selected patients older than 50 to 60 years without an increased risk of death or complications.^{66,67} Furthermore, the rate of reintervention might be lower in the elderly subgroup. Younger patients often present with uni- or bicuspid AV, coupled with higher degrees of AR and a larger aortic annulus,

whereas older patients present predominantly with aortic stenosis.⁶⁶ Secondly, a lower rate of pulmonary homograft dysfunction was observed with increasing age, likely due to the lower inflammatory response in older patients.⁶⁶

A general recommendation is that the Ross procedure should be strongly considered in patients under 50 years of age. Despite limited evidence, we advocate for the use of the Ross procedure in carefully selected patients aged up to 60 years with minimal comorbidities and a life expectancy greater than 15 years. Above 60 years of age, the Ross procedure may be considered selectively, especially in physically active patients who have a family history of longevity and present with a small aortic annulus.

4.2.2 Sex. The Ross procedure represents the ideal solution for women of childbearing age with unreparable AV disease. Cardiac, fetal, and obstetrical risks are not increased after the Ross procedure in pregnant women.⁶⁸ Real-world data, however, report that the majority of adult Ross patients are male.^{6,13,40,69} The possible explanation is that congenital AV disease, the most frequent indication for the Ross procedure in adults,⁷⁰ is observed more often in men than in women.⁷¹

When analysing the Ross procedure outcomes, there were no significant differences in morbidity and mortality observed between men and women.⁷² Women were 4 years older at the time of the operation, in line with other studies.⁷³ They presented more frequently with tricuspid AV configuration and had a higher rate of stenotic or mixed AV disease.⁷² Haemodynamic performance of the autograft was comparable, but the pressure gradient in the homograft was lower in women.⁷² According to some evidence, the long-term freedom from PA failure might be in favour of women, although this result could be attributed to the higher prevalence of aortic stenosis as well as the smaller aortic annulus diameters.^{13,74–76}

4.2.3 Comorbidities. All patient-related factors leading to a significantly reduced life expectancy and/or increased surgical risk warrant consideration of options other than the Ross procedure. Patients with poor LV function, fixed pulmonary hypertension, advanced lung/liver/kidney disease, diabetes with complications, diffuse coronary disease, cancer with poor prognosis, previous mediastinal radiation therapy, and frailty should all be carefully evaluated on a case-by-case basis. The above-mentioned comorbidities may influence the early outcomes of the Ross procedure by several means:

1. Reduced tolerance to longer cross-clamp and cardiopulmonary bypass times
2. High risk of adhesions or fragile tissues, making the procedure technically challenging
3. High risk of postoperative complications, including cardiogenic shock, multiorgan failure, sepsis, prolonged intensive care unit stay, and death

In addition to the higher risk of perioperative mortality and morbidity, these comorbidities also limit the potential benefit of the Ross procedure, given the limited life expectancy of these patients.⁷⁷ In these scenarios, conventional or transcatheter AVR may be beneficial. Rheumatic heart disease, connective tissue disorders, and autoimmune disease are discussed separately below.

Other cardiac and thoracic conditions, such as focal coronary artery disease, mitral valve disease, atrial fibrillation, and aortic arch aneurysms, do not automatically rule out the Ross procedure. Concomitant procedures can be selectively performed in addition to Ross, based on the estimated operative complexity and patient characteristics.

4.2.4 Coronary anatomy. Careful evaluation of coronary anatomy (by means of cardiac computed tomography [CT] or coronary angiography) is mandatory in every Ross candidate. It is noteworthy that patients with congenital AV anomalies have an increased prevalence of anomalous aortic origin of a coronary artery.⁷⁸ There is a wide range of coronary artery anomalies: they may have no impact on the operation, make the operation complex, or even contraindicate it.⁷⁹ The treatment decision depends on the type of anomalous aortic origin and the presence, length, and location of an intramural segment. For instance, in the presence of a retro-aortic course of a left circumflex coronary artery originating from the right coronary sinus (along the noncoronary sinus), care must be taken during aortotomy, dissection of the root, external annuloplasty, and placement of stitches at the proximal suture line.

In contrast, an interarterial course of the right (originating from the left sinus) or left coronary artery/left anterior descending artery (LAD; originating from the right sinus) is associated with higher operative risk, especially during dissection of the root, harvesting the autograft, and placing the stitches at the annuli. Care must be taken during the reimplantation of the coronary ostium; sometimes, an unroofing or more extensive button mobilization might be needed.^{78,80} These coronary artery anomalies are considered relative contraindications for the Ross procedure. An anterior course of a coronary artery across the right ventricular infundibulum (e.g., the right coronary artery originating from the LAD) would inevitably lead to coronary artery damage during autograft harvesting. This situation is considered an absolute contraindication for the Ross procedure.⁷⁹

4.2.5 Aortic valve anatomy. For a long time, the presence of a bicuspid AV was thought to be associated with a poor long-term prognosis after the Ross procedure.^{81,82} A report from the GRR that included a total of 1,277 patients not only showed that patients with bicuspid AV represented the majority of Ross patients in this cohort (70.9%) but also proved that the long-term postoperative autograft failure rate was similar in bicuspid and tricuspid AV morphology.⁸³ Poh et al. observed an 85% freedom from autograft reintervention and/or more than mild AR at 20 years postoperatively in patients with BAV.⁸⁴ Bicuspid AV continues to be the most frequent underlying pathology in patients undergoing the Ross procedure, with no demonstrable difference in early or late outcomes compared to non-bicuspid AVs.

Patients with a unicuspid AV represent a separate entity.⁸⁵ Unicuspid AV is defined by an asymmetrical valvular orifice and a single or no commissural attachment, with or without raphe.⁸⁶ Patients usually present for surgery at a younger age (<35–40 years) and with a more pronounced aortic annular dilation.^{70,85,87,88} These patients represent 27% to 37% of all patients undergoing the Ross procedure in more recent cohorts,^{70,85,89} which is significantly more than previously believed.^{90,91} Nevertheless, neither bicuspid nor unicuspid AV morphology is

associated with a greater autograft failure risk if an appropriate surgical technique is applied.^{23,70,92}

4.2.6 Pulmonary valve anatomy. Significant PV dysfunction on a preoperative echocardiogram is a strict contraindication for the Ross procedure. Although trace to mild central pulmonary regurgitation is almost always present and has no impact on long-term performance in the aortic position, more severe regurgitation or an eccentric jet may preclude the Ross procedure. Small commissural jets are also often observed and bear little consequence on valve performance. Because transthoracic and transoesophageal echocardiography (TOE) have inherent limitations in evaluation of the PV, direct epicardial ultrasound before initiation of cardiopulmonary bypass may provide additional information regarding PV anatomy and function.⁹³ In addition to the echocardiographic examination, careful intraoperative visual valve inspection is equally important. The presence of large cusp fenestrations, especially in the peri-commissural area, is a contraindication for the Ross procedure because the remaining tissue strands can rupture once they are exposed to systemic pressures.⁷⁹ Visual inspection can also reveal a congenital PV anomaly. The estimated prevalence of bicuspid and quadricuspid PVs in the population is 0.1% and 0.2%.⁹⁴ In patients with congenital AV anomalies, the incidence is somewhat higher, around 0.5%.⁷⁹ The decision to utilize a bicuspid or quadricuspid PV as an autograft for the Ross procedure is generally not recommended, although it is technically possible. It has shown acceptable mid-term results in small series and case reports.^{79,95}

If there is a significant discrepancy in size between the aortic and pulmonary annuli (>2 mm), the Ross procedure may still be performed. Unless it is addressed, the larger size of the aortic annulus may lead to a higher risk of autograft failure postoperatively.⁹⁶ Therefore, some form of reduction/stabilization must be added to reduce this risk (see Chapter 5).^{24,97} A larger pulmonary annulus usually does not impose a greater risk, and the valve functions properly in the long run postoperatively without any additional intraoperative interventions.

4.3 Indications for the Ross procedure

The established indications for AVR due to AV disease (stenosis, regurgitation, or mixed disease) are well documented in the literature. Nevertheless, the role of the Ross operation remains inadequately emphasized in both European and American guidelines.^{1,2} Given the importance of the indications for the Ross procedure, a comprehensive anatomical assessment of both isolated and mixed AV pathologies is crucial in determining the feasibility and appropriateness of the Ross procedure in each patient.

4.3.1 Aortic valve stenosis. Both isolated AV stenosis and mixed AV disease with predominant stenosis, particularly when associated with a non-dilated aortic annulus (below 27 mm), present favourable anatomical conditions, i.e. "ideal anatomical substrate" for the Ross procedure.^{17,98} These conditions are associated with favourable long-term results irrespective of AV morphology, because no differences in autograft reintervention were observed between tri-, bi-, and unicuspid AV morphology.^{70,92,99}

4.3.2 Aortic valve regurgitation. Regurgitant AV, including bicuspid valves, should primarily be repaired or spared if durable results can be expected.^{1,2,100} If repair is not possible or judged to be

of limited durability, the Ross procedure is a valuable option for non-elderly adults. A similar late survival benefit is observed in patients with AV stenosis or regurgitation.^{84,89} However, there are concerns about the durability of the Ross procedure when treating patients with isolated AR, a dilated aortic annulus (≥ 27 mm), and aortic/pulmonary size mismatch.^{13,33,41,43,74} These are perceived as risk factors for autograft failure and the subsequent need for reintervention. Nevertheless, with tailored surgical techniques, comparable long-term results have been demonstrated.^{89,92}

The above-mentioned anatomical conditions represent a less optimal anatomical substrate for the Ross procedure. However, these patients may still benefit from the Ross procedure.^{17,98} Importantly, a tailored free-standing root replacement technique with autograft reinforcement or prosthetic/autologous inclusion technique, as well as application of strict blood pressure control, is crucial in these patients to achieve durable late results.^{89,92}

4.3.3 Infective endocarditis. The choice of valve substitute in patients with infective endocarditis is always challenging, especially in young patients and intravenous drug users. The PA is living tissue with good resistance to infection, which is not the case for the pulmonary homograft, for which infection rates are higher. The number of publications in which the Ross procedure was employed in the setting of active infective endocarditis is limited. Nevertheless, mid-term follow-up showed low rates of reinfection, good autograft durability, and good patient survival.^{55,101–103} Possible candidates are young and otherwise healthy individuals with native/prosthetic AV endocarditis and no or limited annular involvement. Other patient-specific circumstances, such as contraindication to vitamin K antagonists or active intravenous drug abuse, are considered. The use of prosthetic material should be limited or completely avoided. The Ross procedure for infective endocarditis is a complex operation and should be performed in experienced centres only.

4.3.4 Rheumatic involvement of the aortic valve. Although rheumatic heart disease is rarely seen in developed countries, it remains the leading cause of valve disease in developing countries. It usually manifests clinically in young adulthood. Use of a PA in patients with rheumatic involvement of the AV remains a matter of debate because of concerns that the autoimmune disease would potentially affect a PA in the aortic position.

In the last 3 decades, numerous researchers have examined the role of the Ross procedure in patients with rheumatic valve disease, including meta-analyses and multicentre studies.^{104–107} The results are to some extent contradictory—older studies showed suboptimal results, with increased PA failure rates; recent studies report improved long-term outcomes similar to those of non-rheumatic patients, but the number of patients and studies is limited. Despite this recent trend, there is still not enough evidence generally to recommend the Ross procedure in patients with rheumatic heart disease. It may be considered in selected young individuals, but the possibility of the limited durability of the PA must be accepted. Moreover, suboptimal results of other possible valve substitutes must be taken into consideration, especially in developing countries with limited access to medical care. The long-term use of penicillin G prophylaxis is recommended, but its specific effectiveness has not been formally studied.

4.3.5 Connective tissue disorders. Marfan syndrome (MFS) is the most common hereditary syndromic aortopathy, leading to

aortic aneurysm formation and the subsequent high risk of acute aortic syndrome.¹⁰⁸ MFS, along with other syndromic aortopathies such as Loeys-Dietz syndrome and Ehlers-Danlos syndrome, represents a contraindication to the Ross procedure because the underlying media degeneration may also affect the wall of the pulmonary artery.^{109,110} In most instances, the underlying AV pathology in patients with MFS is the root dilation and the subsequent AR; valve-sparing root replacement is usually the procedure of choice.¹¹¹

4.3.6 Autoimmune diseases. Evidence describing the utilization of the Ross procedure in patients with autoimmune diseases (e. g., rheumatoid arthritis, Takayasu arteritis, lupus erythematosus) is scarce. Performing the intervention during the active phase of the disease is contraindicated.¹⁰⁵ There is a risk of early PA failure due to

candidate. Evaluation of the coronary anatomy is mandatory in all patients by means of cardiac CT or coronary angiography. Cardiac CT is helpful to visualize the coronary anatomy in three dimensions and the association with other cardiac structures, to rule out coronary artery disease, provide additional information regarding aortic and pulmonary annulus size, and look for eventual thoracic aortic pathology (not infrequent in this patient population). Left heart catheterization, including a coronary angiogram, should be performed to identify coronary artery disease in patients at risk for atherosclerosis.¹¹⁴ Angiographic visualization of the first septal branches is helpful for surgical planning for harvesting the PA. Other tests, such as cardiac magnetic resonance imaging, may be employed as part of the indication process in accordance with guidelines,² but are not specifically necessary before the Ross procedure.

Consensus Statement Table 2. Patient Selection
• The Ross procedure should be considered in adult patients aged less than 60 years with minimal comorbidities, following a shared decision-making process.
• In women of childbearing age requiring AVR, the Ross procedure is the preferred option.
• Coronary anomalies should be approached with caution in patients being considered for the Ross procedure.
• The optimal indication for the Ross procedure is AS or mixed disease with predominant stenosis, particularly when associated with a non-dilated aortic annulus.
• Patients with pure AR who are not eligible for AVR are good candidates for the Ross procedure if the appropriate surgical technique is used.
• The Ross procedure may be considered in selected patients with rheumatic heart disease or with infective endocarditis.
• The Ross procedure should be avoided in patients with genetically confirmed syndromic aortopathy, such as Marfan syndrome.
• The Ross procedure should be avoided in patients with autoimmune disorders.

Abbreviations: AR: aortic regurgitation; AS: aortic stenosis; AVR: aortic valve repair; AVR: aortic valve replacement.

autoimmune deposits, which can cause cusp retraction and AR. Nevertheless, autoimmune diseases are not an absolute contraindication. Indeed, most often these patients are young, female, of child-bearing age, and with a small aortic annulus. As a result, the haemodynamic and pregnancy-related benefits of the Ross procedure should be considered against the reduced durability of the PA.

If the Ross procedure is considered in such patients, a positron emission tomography-computed tomography scan should be performed preoperatively to rule out an active phase of the disease. If a quiescent disease state is reached, the operation can be undertaken, followed by long-term rheumatology follow-up. Limited single-centre evidence has shown optimistic mid-term outcomes in a very small cohort of 16 patients with up to 10 years of follow-up.⁷⁹ In summary, the decision to utilize the Ross procedure in a patient with an autoimmune disease must be made on an individual basis and cannot be generally recommended.

4.4 Practical aspects of preoperative patient selection

Apart from the standard preoperative evaluation and calculation of perioperative mortality risk, the Ross procedure requires a few specifics, particularly in terms of imaging.^{112,113} Transthoracic and TOE are needed to evaluate anatomical defects and ventricular and valvular function. A detailed study of the aortic and pulmonary roots, including dimensions, anatomy, and valve function, is crucial in a Ross

5. SURGICAL TECHNIQUES OF THE ROSS PROCEDURE

The original technique of implanting a PA, described by Donald Ross, was a subcoronary (SC) technique.^{14,115} The PA is trimmed, leaving just a small amount of tissue along the PV cusp attachment. The resulting scalloped PV is then implanted in the aortic position within the native aortic root.

Long-term outcomes of the SC technique are well documented in the GRR.¹¹⁶ Preservation of the aortic root may protect the PA from dilatation. On the other hand, any pathological changes in aortic root geometry can lead to PA failure. This technique can be challenging or even contraindicated in patients with congenital malformations of the aortic valve (e.g., type 0 bicuspid AV) because PA inclusion inside the asymmetric aortic annulus may lead to distortion of valvular geometry. Very close positioning of the coronary arteries to the aortic annulus, as may be the case in bicuspid AV, also makes this technique more challenging and sometimes technically impossible. Although the results of SC implant of PA are favourable, this technique has gradually fallen out of favour.

Technical complexity and the limited reproducibility of the SC technique led to the development of the root replacement (RR) technique, also called “free-standing root replacement”. This technique is more widely applicable because of the limited influence of native aortic root asymmetry on the feasibility of the

Ross procedure. The initial enthusiasm for the RR technique as originally performed eventually waned due to the increased incidence of observed PA dilatation in the mid-term, leading to AR and subsequent reoperation.^{117–119} Disturbing results were reported by Klieverik et al., with freedom from reoperation of only 69% at 13 years.¹⁸ Understanding the underlying causes of PA failure after the Ross procedure led to a re-examination of the RR technical details and improvement of the surgical technique, aiming at optimizing long-term outcomes. Addition of reinforcement of the PA at the level of the basal ring and the sinotubular junction (STJ) to the standard RR technique, together with addressing the mismatch between the pulmonary and the aortic annulus, was subsequently termed reinforced root replacement (RR+R).^{96,120,121}

The most extensive data in terms of RR+R effectiveness comes from the GRR. The root replacement without reinforcement was associated with a 6-fold increase in reoperation rates compared to RR+R due to progression of AR.²⁵ Also, the presence of pure AR preoperatively was identified as a predictor of reoperation.¹²² Therefore, in root replacement, the reinforced technique has become the procedure of choice.

Over the last 2 decades, other technical modifications have been described to provide external support to the autograft and to reduce the risk of PA dilatation and failure, especially in patients with pure AR and/or dilated aortic annulus and aorta. These include a tailored free-standing root replacement, native aortic root inclusion using the autologous support technique and inclusion of the PA within a prosthetic Dacron tube graft-prosthetic inclusion technique.^{22,89,123}

5.1 Tailored free-standing root replacement

The original RR technique has been modified over time to prevent PA dilatation at all levels (aortic annulus, sinuses of Valsalva, and STJ).^{23,89} The main technical steps of the tailored free-standing root replacement technique are the following:

1. The pulmonary artery is transected about 1 cm to 2 cm proximal to the bifurcation, and the valve is evaluated for feasibility. Large fenestrations, retraction, or significant asymmetry render the valve unsuitable for the Ross procedure.
2. The PA is harvested so that only 2 mm to 3 mm of the infundibular muscle are left along the cusp attachments proximally. The harvest usually starts 5 mm below the PV to prevent cusp damage. The muscle is trimmed to the desired length after the harvest is finished. Special attention must be drawn to the left main stem as well as to the LAD with its first septal branch.
3. The aortic valve is excised, and coronary buttons are mobilized.
4. In case of AR and/or aortic annulus dilation (> 26 mm), deep aortic root dissection followed by external aortic annuloplasty is employed (see below).
5. The PA is implanted in a subannular position within the LVOT so that the native aortic annulus may provide additional support to the PV in the aortic position. In patients with congenital malformation of the aortic valve, there is often an asymmetry in commissural orientation and nadir depth. Care must be taken to implant the PA symmetrically (in 120° – 120° – 120° orientation) and in a single horizontal plane (**Figure 2**) with an ascent at the right-non commissure to avoid damage of the conduction system. If some asymmetry of pulmonary valve commissural orientation is present, it should be

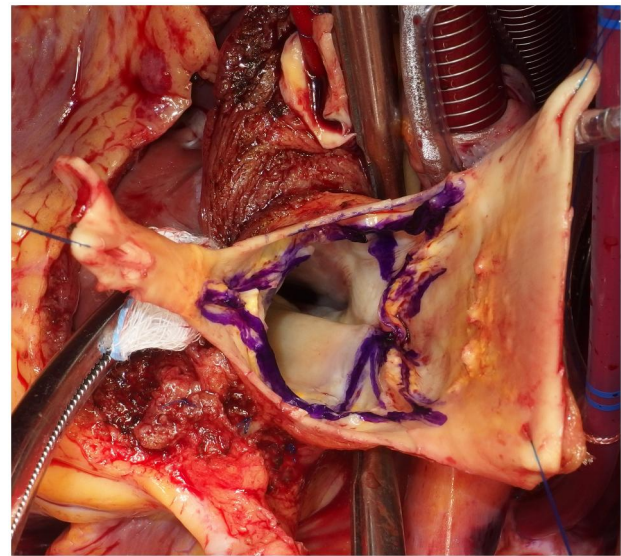


Figure 2. Tailored Free-standing Root Replacement: Deep and Symmetric implantation of the Autograft

respected when the PA is being implanted. The proximal anastomosis can be accomplished with interrupted or running sutures. In patients with no indication for external annuloplasty, a Dacron strip can be incorporated into the proximal anastomosis to reinforce the annulus.

6. Subsequently, the coronary arteries are reimplemented into the neo-aortic root, within their respective sinuses.
7. The pulmonary homograft is implanted into the pulmonary position.
8. The PA is then cut distally, leaving 2 mm to 3 mm above the tips of the commissures to prevent dilatation of the redundant pulmonary artery. The distal anastomosis is performed at the level of the autograft STJ to stabilize the commissures and prevent STJ dilation, which could lead to functional AR. For this reason, we advocate a liberal threshold for replacement of the ascending aorta using a short interposition graft or reduction aortoplasty in mild-to-moderate dilatation of the ascending aorta (above 32–36 mm) or patients with a mismatch between the STJ of the autograft and the ascending aorta. The diameter of the tubular graft is 26 mm or 28 mm in most patients.
9. The remnant native aortic root may be used to prevent dilation of the PA at the level of the sinuses of Valsalva ("loose jacket"). This is appropriate, especially in patients with AR and a dilated root.^{92,124} (**Figure 3**).

5.1.1 External aortic annuloplasty. Dilatation of the aortic annulus has been recognized as an important risk factor for the recurrence of AR after the Ross procedure. Two reasons may play a role: inadequate reduction of the aortic annulus during the index procedure or recurrent dilatation of the unsupported annulus during follow-up.

Different techniques have been proposed to reduce and stabilize the aortic annulus in the Ross procedure. These include subcommissural annuloplasty, partial Dacron/pericardial strip, suture annuloplasty, extra-aortic annuloplasty, or incorporation of a Dacron/pericardial strip into the proximal suture line. The most consistent data regarding the effectiveness of the

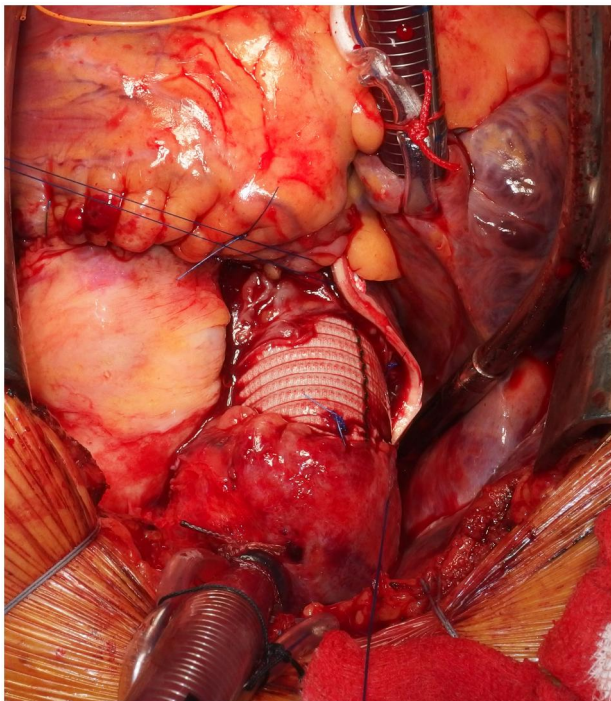


Figure 3. Tailored Free-standing Root Replacement: Loose Jacket Technique and Supracoronary Aortic Replacement

annuloplasty comes from the GRR, where the annulus was stabilized with the Dacron strip in most patients. During the mid-term follow-up, the annuloplasty had a significant protective effect against PA failure.²⁵ Unfortunately, in the latest GRR report, the long-term effect of the annuloplasty was not evaluated.¹³ No studies comparing the efficacy of different annuloplasty strategies in the Ross procedure have yet been published, and the evidence is limited to the level of expert opinion.

In the last 2 decades, there has been a significant evolution of AV repair techniques, including aortic annulus stabilization with external circumferential annuloplasty as the most effective and durable technique.^{125,126} Extrapolation of these techniques into the Ross procedure has been described as well. External aortic annuloplasty using either a dedicated expansible aortic ring (Extra-Aortic; CORONEO Inc., Montreal, Quebec, Canada) or a short ring of a Dacron vascular prosthesis has become a standardized technique with predictable results.^{24,127,128}

The implant process begins with a deep aortic root dissection down to the level of the virtual basal ring.^{127,128} Five or six polyester pledgeted mattress sutures are placed from inside the LVOT to the outside of the aortic root, approximately 5 mm below the plane of the basal ring, to avoid any contact between the polyester pledgets and the PA tissue (Figure 4). Later, they are used to implant the external ring before the PA is implanted. Care is taken to avoid the conduction tissue around the membranous septum by raising that suture 2 mm or placing the stitch outside of the aorta. The sizing of the ring is based on the diameter of the PA annulus measured with TOE and intraoperatively after its harvesting using a Hegar dilator. The aim is to reduce and stabilize the aortic annulus to create an optimal landing zone for the PA. To account for the thickness of the tissue between the ring and the inner diameter of the PA, the following formula is recommended for sizing: Size of the Dacron ring = PA annulus diameter + 6 mm. Effective, predictable, and durable

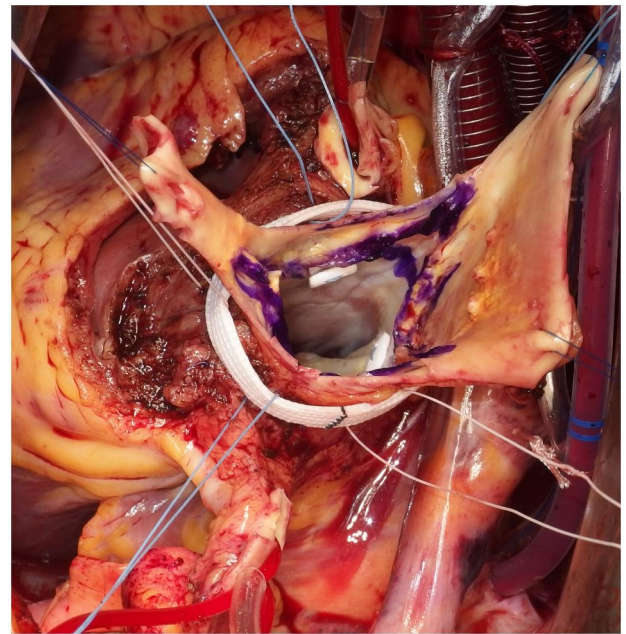


Figure 4. Tailored Free-standing Root Replacement: Extra-aortic Ring Annuloplasty

reduction of annulus size by this technique has been observed at the mid-term follow-up.²⁴

5.2 Autologous inclusion technique

The technique of placing the PA inside the patient's own aortic root was championed by Skillington in the late 1990s.¹²⁹ The rationale for this technique is to avoid late dilation of the PA in both a transverse and a longitudinal direction, which could lead to AR and failure of the Ross procedure.

An advantage of this technique compared to prosthetic root inclusion is that it reduces, and in some cases abolishes, the use of prosthetic material in the aortic root. There is a reduction in neo-aortic cusp stress as the neo-aortic valve cusps are opening and closing within an expansile chamber rather than within a rigid prosthetic root. In comparison to performing the unsupported RR method, haemostasis is improved at the annulus level, because the PA root is enclosed within the aortic root, which obviates bleeding at the annular level. Anchoring the distal autograft suture line at the STJ within the lower ascending aortic remnant prevents not only transverse expansion but also longitudinal expansion of the autograft root. This technique has demonstrated excellent durability and function of the autograft for up to 30 years.¹³⁰

A disadvantage of this technique is its relative complexity compared to other techniques, which limits its reproducibility. This relates mainly to ensuring that the coronary arteries are not kinked at the exit points from the aortic root via the fenestrations created by excision of the coronary ostia. Alternatively, coronary buttons can be reimplanted without mobilizing them, as proposed by Ozturk et al.¹³¹ An additional requirement, which is similar to other techniques, is that an annuloplasty is necessary if the aortic annulus is enlarged above 26 mm. Also, if the aortic sinuses are enlarged, this is generally reduced via excision of the upper portion of the noncoronary sinus via a

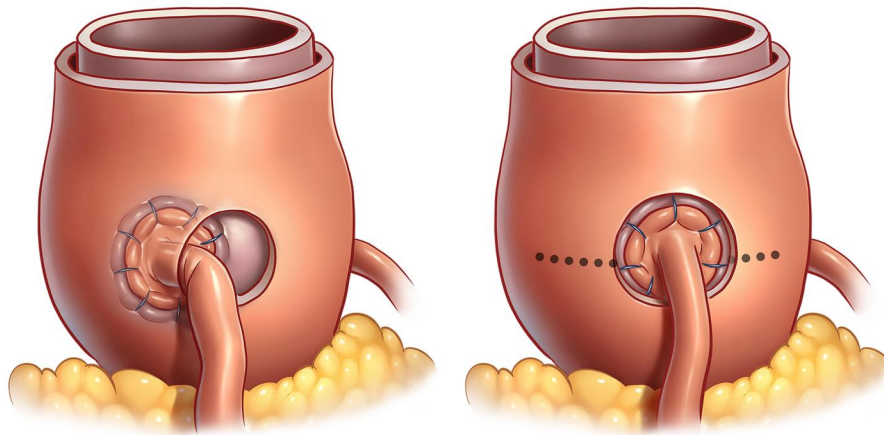


Figure 5. Autologous Inclusion Technique: Incision Around the Left Coronary Ostium to Prevent Kinking

triangular excision, so that the diameter of the STJ of the autograft is similar to that of the altered aortic STJ diameter. Similar to other techniques, if the ascending aorta is enlarged, this requires either ascending aortic replacement with an interposition Dacron graft, or in less severe cases, reduction aortoplasty.

The main technical steps of the autologous inclusion technique are the following:

1. A transverse aortotomy and aortic transection are performed 1 cm above the STJ; the aortic valve is inspected and excised, and the aortic annulus is completely decalcified.
2. The internal aortic annulus is measured, and the valve morphology is noted. The STJ diameter is measured from either the CT scan or the echocardiogram.
3. A vertical extension of the transverse aortotomy incision is performed down into the noncoronary sinus, all the way to the annulus. This step opens up the aortic root and provides excellent exposure for subsequent manoeuvres.
4. The distal main pulmonary artery is transected, and the proximal main pulmonary artery is dissected clear of the surrounding tissues within the adventitial plane. The PV is evaluated and eventually deemed acceptable for the Ross procedure.
5. The aortic annuloplasty is performed (this is required in 60%-70% of cases). The preferred technique is a partial external Dacron ring annuloplasty based on the most enlarged aortic sinus. The internal aortic annulus diameter is reduced to 24 mm to 26 mm in men and to 22 mm to 24 mm in women. Annuloplasty using horizontal mattress non-pledgeted sutures are placed from inside to outside around the portion of the aortic annulus to be narrowed, passed through a 4-mm wide Dacron band. If the partial annuloplasty ring encompasses the left coronary sinus, the left coronary artery needs to be detached at this stage to complete the narrowing of the aortic root. The external annuloplasty based on the most enlarged sinus brings the aortic valve commissures to be either equidistant or near equidistant.
6. The PA is then fully harvested, and the base is trimmed to within 4 mm of the valve annulus.
7. The pulmonary homograft is sewn into the right ventricular outflow tract (RVOT) once haemostasis of the PA bed has been completed. The right ventricle is de-aired prior to completing the suture line.
8. Small holes are created in the sinuses of the PA for later anastomoses of the coronary ostia, using a 4-mm punch. The positions of the holes are based on the locations of the coronary ostia in the native aortic root.
9. The proximal autograft suture line is performed in a scalloped fashion using simple interrupted sutures, lining up the PA commissures with the original aortic commissures (which have been rendered more equidistant via placement of the eccentric annuloplasty band).
10. The coronary ostia buttons are now fully detached; they are delivered through the resultant fenestrations and anastomosed to the previously created holes in the PA sinuses. The fenestration created by excision of the left coronary sinus is enlarged by approximately 2 mm in the medial and lateral directions on each side to allow for the unrestricted exit of the left coronary artery once the PA is pressurized later (**Figure 5**).
11. The vertical extension of the aortotomy incision down into the noncoronary sinus has featured excision of a triangular width of tissue with the base of the triangle at the STJ; this tissue is now sewn up directly with continuous suture, reducing the size of the aortic root, which now aligns with the STJ of the autograft (**Figure 6**).
12. If the ascending aortic diameter at the level of the aortotomy is larger than the STJ, it is reduced either via vertical reduction aortoplasty or via replacement of the ascending aorta with a Dacron graft.
13. Finally, the height of the autograft cylinder is trimmed so that it is the same height as the lower ascending aortic remnant. Then the PA is anastomosed end-to-end to the patient's distal ascending aorta. For the portion of this suture line immediately cephalad to the left coronary ostial anastomosis, the lower ascending aortic remnant is included in this suture line, to ensure that the outer cylinder (aortic remnant) does not slip down and occlude the left coronary artery (**Figure 7**).
14. After release of the aortic cross-clamp, with the heart beating, both coronary anastomoses are checked to ensure that the coronary ostia are free and unencumbered, not kinked.
15. While the heart is rewarming, 3 or 4 figure-of-eight sutures are placed anteriorly above the right coronary anastomosis, to hitch up the anterior ascending aortic remnant to the distal autograft suture line, meaning that the autograft is now fully encompassed within the patient's aortic root.

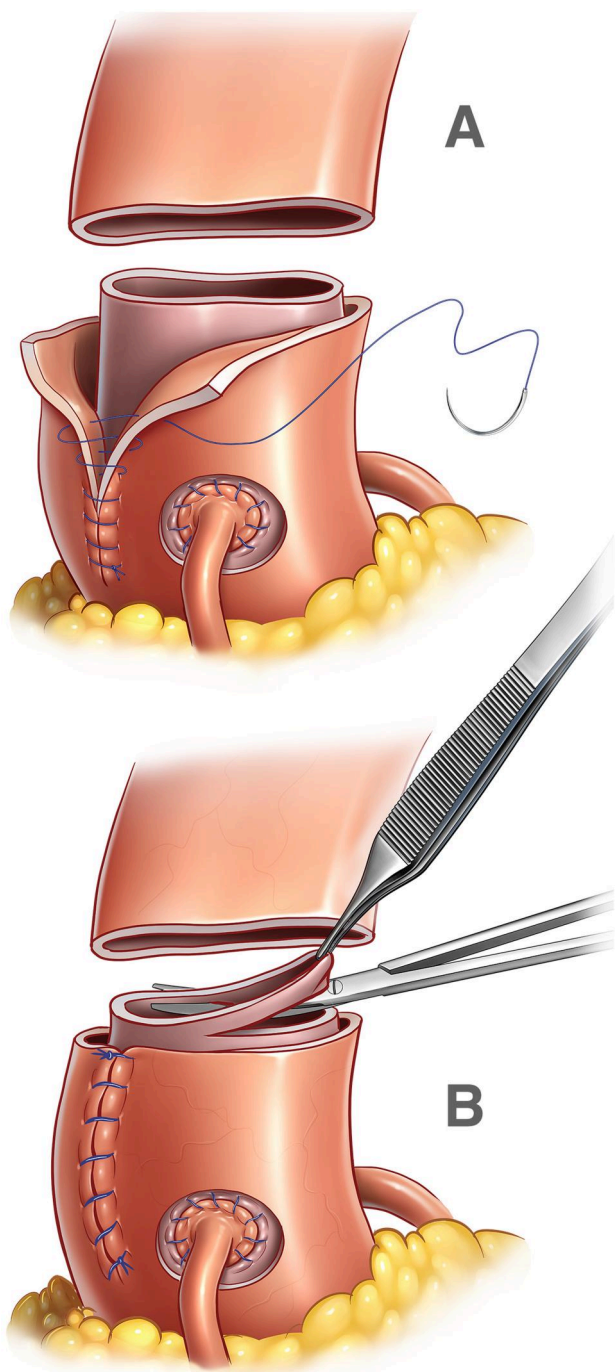


Figure 6. Autologous Inclusion Technique: Vertical Continuous Suture of Native Aortic Root Remnant (A and B)

5.3 Prosthetic inclusion technique

The Ross procedure featuring the inclusion of the PA in a Dacron prosthesis was developed in the early 2000s with the rationale to protect the PA from progressive dilatation observed after RR.^{18,132-134} The prosthetic inclusion technique is an alternative to autologous inclusion if the aortic root size does not correspond with the size of the PA. Prosthetic inclusion provides a full and calibrated support of the PA from the basal ring to the STJ, while the distal ascending aorta may be replaced with the same prosthesis if indicated. A disadvantage of this technique is

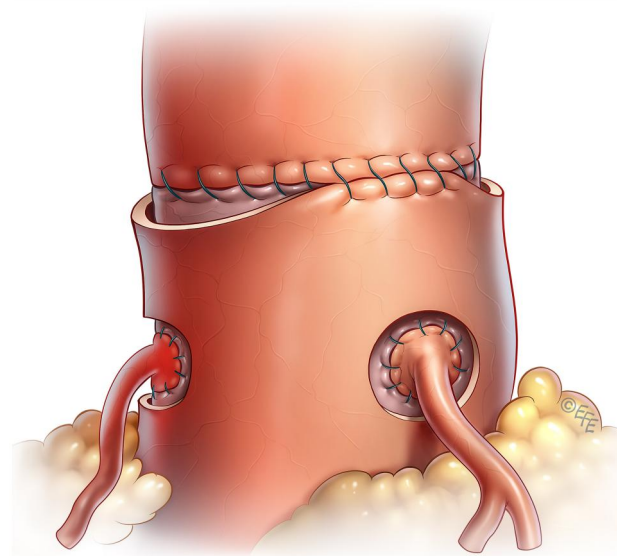


Figure 7. Autologous Inclusion Technique: Distal Suture Line and Fixation of Native Aortic Root Above The Coronaries

the placement of the PA in a less compliant environment, which may have a negative impact on valve function in the long term. However, the experience with the AV reimplantation technique (David procedure) has demonstrated that valve function can be well preserved for up to 2 decades inside a Dacron graft.^{135,136} Whereas the literature provides more robust and long-term data on RR and RR+R, several smaller series have shown favourable mid-term results with the prosthetic inclusion technique.^{134,137,138} Despite its apparent complexity, the prosthetic inclusion technique is safe and durable for at least 1 decade postoperatively, according to current knowledge.¹³⁴ The Dacron graft ensures the stability of the PA annulus and the root dimensions, which could eventually lead to a more durable long-term valve function compared with unsupported RR techniques, particularly in selected patients with a dilated aortic annulus and aorta.^{134,138,139}

Different prosthetic inclusion techniques have been described, depending on the type of Dacron graft used (straight vs Valsalva graft) and on the way the PA inclusion is managed (cylinder vs scalloped).

The main technical steps of the prosthetic inclusion techniques are the following:

1. After an aortotomy 1 cm above the STJ, the AV is excised, and the aortic annulus is completely decalcified.
2. The PA is harvested in the same manner as that used for the RR technique.
3. The aortic root is prepared by opening the non-coronary sinus longitudinally, followed by excision of the right and left coronary buttons.
4. The size of the Dacron graft is determined by the height of the PA measured from inflow to the STJ or by the diameter of the inflow measured with a valve sizer plus 2 mm.^{97,139} The graft sizes usually range between 26 and 30 mm, the 28-mm size being the one most frequently used.
5. The PA inflow is not scalloped but trimmed to make a planar circumference, leaving 4 mm of muscle below the valve nadirs. The tips of the PV commissures are sutured to the neo-STJ, respecting the height of the commissure and their 120°

orientation (**Figure 8A**). The PA inflow is then sutured to the inflow of the Dacron graft with a circumferential running suture (**Figure 8B**). If a Valsalva graft is chosen, the proximal skirt is cut to leave only 2 mm to 3 mm below the neo-root portion of the graft. In the scalloped technique, the PA sinus walls are removed at this stage, and a running suture is used to fix the remnant of the sinus wall to the graft, as is done in the reimplantation technique. Valve configuration and closure can be evaluated by flushing the PA root with saline solution.

6. The PA within the prosthesis is then fixed to the aortic annulus following a horizontal plane with 3 running sutures (**Figure 9**); interrupted sutures can be used, too, based on the surgeon's preference. The basal portion of the membranous septum is left intact to avoid bundle branch block. The suture attachments of the valve commissures to the graft can be readjusted at this stage to acquire optimal valve geometry.
7. If the cylinder technique is chosen for PA inclusion, the PA outflow is sutured to the graft with a running suture at the level of the STJ (**Figure 10**). Left and right coronary ostia are reimplanted into the graft, including both the Dacron and the

PA sinus wall in the suture line. If the scalloped technique is chosen, coronaries are anastomosed only to the Dacron graft.

8. The pulmonary homograft is implanted in a standard fashion.
9. The length of the Dacron graft is trimmed, and the distal aortic anastomosis is completed at the level needed for aortic replacement.

An alternative to Dacron reinforcement is the use of a polyethylene terephthalate microporous mesh in the form of a Personalised External Aortic Root Support (PEARS) prosthesis.¹⁴⁰ This technique was initially described as an external support of the aortic root to prevent its dilation in patients with MFS or other connective tissue disorders.¹⁴¹ The graft is manufactured individually based on a 3-dimensional model of the aortic root from a preoperative CT scan. This concept was recently adopted in the Ross procedure with the rationale to provide personalized external support to the PA (based on the 3-dimensional CT model of the pulmonary root) and thus prevent its dilation and failure in the long term. It is implanted around the PA with 2 apertures for the coronary buttons.¹⁴² In case of ascending aortic dilation, a concomitant reduction aortoplasty is needed. The

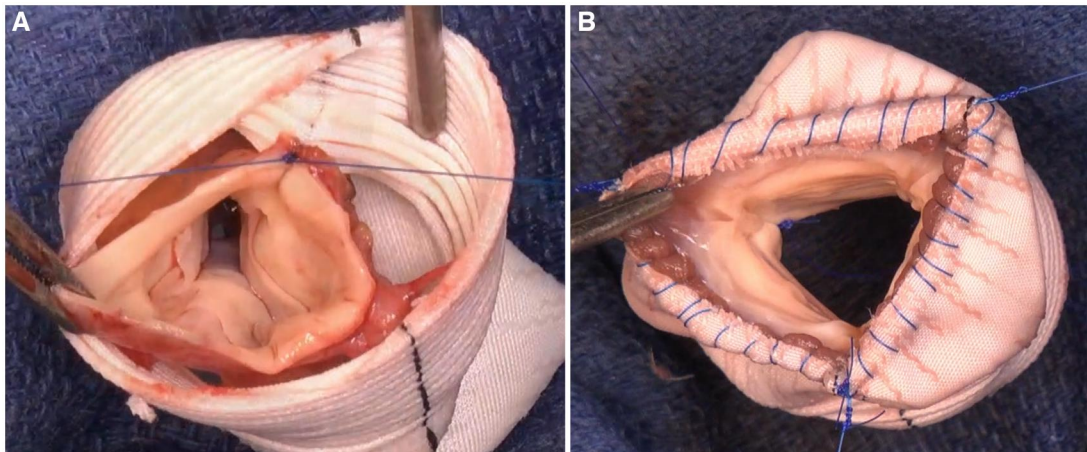


Figure 8. Prosthetic Inclusion Technique: Fixation of the Autograft at the Distal (A) and Proximal (B) Levels

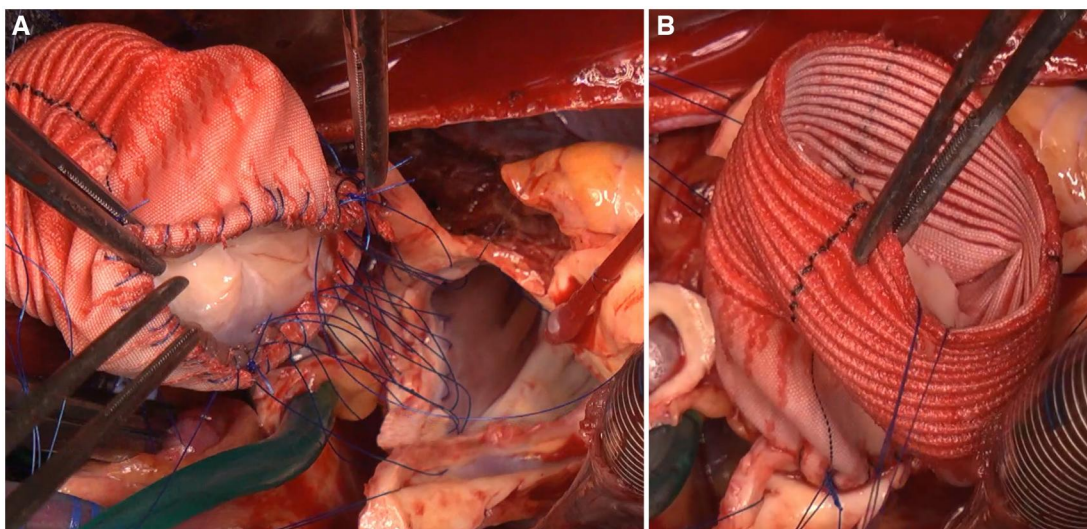


Figure 9. Prosthetic Inclusion Technique: Autograft Proximal Suture Line (A) and Reimplantation of Coronaries (B)

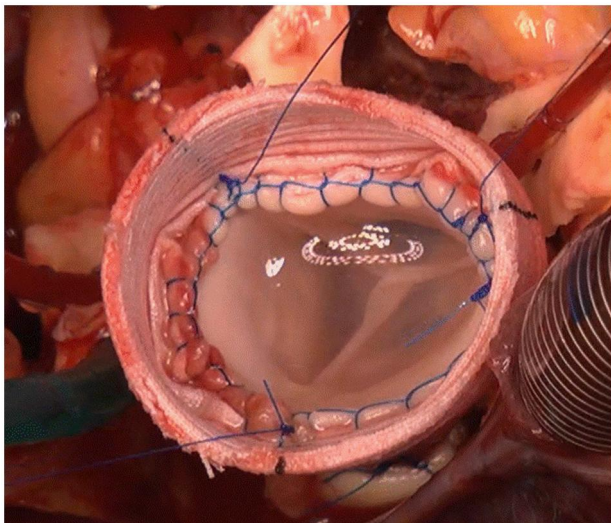


Figure 10. Prosthetic Inclusion Technique: Continuous Suture at the Autograft Sinotubular Junction

technique is referred to as “Ross-PEARS”. Thus far, limited data are available regarding the mid- to long-term results of this technique.

5.4 Postoperative TOE evaluation and when to reclamp

Using TOE to evaluate the outcome in the operating room right after the procedure is an essential part of the Ross procedure. The aim is to assess the function of the PA and the homograft under physiological loading conditions (patient completely off bypass, appropriate LV filling, systemic normotension, and physiological heart rate).

Assuming there is no apparent technical issue (high quality of PV, symmetrical PA implant, annuloplasty performed if necessary), the function of the PA is optimal in most patients. Residual central trace or mild AR is not considered a pathological finding because it remains stable in the long term and does not increase the risk of reoperation.

However, more than mild residual AR is of significant concern. Although we do not have enough scientific evidence (like in AV repair), more than mild residual AR is considered a predictor of early autograft failure and reoperation. Eccentric regurgitant jets are particularly worrisome, often indicating an underlying technical issue (e.g., asymmetric implantation of autograft at the level of proximal or distal anastomosis, leading to cusp prolapse).

The role of an experienced echocardiographer is to reveal the mechanism of residual AR and, together with the cardiac surgeon, to assess possible solutions. If it is not possible to identify an apparent cause of residual AR that could be eliminated by surgical reintervention (e.g., reduction of STJ) and the degree of regurgitation is greater than moderate, it is necessary to proceed with aortic valve/root replacement. To conclude, when dealing with residual AR in the Ross procedure, we apply the same criteria as in AV repair.

Other possible complications requiring reclamping, such as bleeding from a proximal anastomosis or coronary kinking, require an approach similar to that of other aortic root procedures. This observation underlines the fact that the surgical team

and echocardiographers involved in the Ross program must be experienced in aortic root surgery, as suggested in Chapter 8.

5.5 Different substitutes for pulmonary root replacement

After meticulous harvesting of the autograft, the RVOT and pulmonary trunk must be reconstructed. Homografts and xenografts are straightforward solutions because they best imitate the native PA shape. They are easier to use compared to valved conduits, which have a different shape and require adjustments prior to being implanted or the use of additional patch material to reconstruct the RVOT. After PA harvesting, the autograft bed must be meticulously checked for potential bleeding. Usually, the cardioplegia is given at this phase, and bleeding is controlled by electrocautery, oversewing, and/or addition of glue. The graft implantation starts with the distal anastomosis, followed by the proximal anastomosis. Care must be taken to choose the right length of the graft. While too short grafts may lead to tension, bleeding, and rupture, too long grafts may kink and cause stenosis. Adjusting the homograft length based on the post-harvest PA length (before any trimming) can help prevent those complications. Furthermore, utmost attention must be paid to not twisting the graft.

Since the introduction of the initial concept of the Ross procedure, pulmonary homografts have been used to replace the harvested autograft.¹¹⁵ Although several alternatives have been proposed, pulmonary homografts represent the optimal substitute for right-sided reconstruction in the Ross procedure in adults. The favourable results are probably related to the specific anatomical shape of the RVOT, which needs to be resembled by any implant used, and by the low-flow and infection-prone location of the implant, making conventional bioprotheses not the best solution. Despite excellent results, various risk factors have been reported to be associated with a higher incidence of homograft dysfunction, including young recipient age, young donor age, small homograft diameter, and male gender (maybe related to oversizing in female patients).¹⁴³ Failures occurring early after surgery may potentially be associated with an immunological response.¹⁴⁴ Stentless xenografts represent a good alternative to pulmonary homografts, although contradictory results have been observed. Kanter et al. have shown superior outcomes of xenografts compared to homografts; these procedures, however, were performed in the paediatric population.¹⁴⁵ Hechadi et al. observed comparable outcomes of both alternatives despite somewhat more pronounced calcification of the xenograft wall.¹⁴⁶ Miskovic et al. reported a trend towards strictures at the proximal suture line.¹⁴⁷ A pericardial sleeve sutured between the RVOT and xenograft might overcome this issue.¹⁴⁸ Even if decellularized xenografts were used, the results were not satisfactory.¹⁴⁹ In a meta-analysis by Van den Eynde et al., the authors concluded that xenografts may be used as an acceptable alternative for RVOT reconstruction in the Ross procedure, although it may be associated with a higher rate of reinterventions and PV dysfunction.^{150,151}

Whereas homografts are the best available solution, they are still at risk of degeneration, which may cause reintervention or reoperation. It is important to use an appropriately sized graft because some shrinkage may be expected. Oversizing of the homograft (>27 mm) can improve durability. Furthermore, the quality of the homograft can significantly impact its durability. Risk factors for early degeneration may include prolonged warm ischaemia prior to harvesting.

Oeser et al. showed inferior results of grafts harvested after prolonged warm ischaemia compared to grafts harvested directly during the transplant procedure and stored immediately.¹⁴³

6.1.1 Incidence of autograft failure. Autograft failure after the Ross procedure has been reported in various studies, with rates varying depending on the patient population, surgical

Consensus Statement Table 3. Surgical technique for the Ross procedure

• Tailored free-standing root replacement, native inclusion, and prosthetic inclusion are currently the most appropriate surgical techniques to ensure the best long-term results of the Ross procedure.
• Annuloplasty should be performed in all patients with pure or predominant AR and in those with a dilated aortic annulus ≥ 27 mm or a mismatch between the aortic and pulmonary annuli.
• In patients with a mismatch between the sinotubular junction of the autograft and the ascending aorta, an interposition graft or reduction aortoplasty should be performed.
• A cryopreserved pulmonary homograft is the gold standard for right-side reconstruction in the Ross operation.
• A xenograft valve is an alternative to the pulmonary homograft, but its expected durability is inferior.
• Decellularized fresh or cryopreserved homografts may offer better durability compared to standard homografts.

Abbreviation: AR, aortic regurgitation.

Decellularization of pulmonary homografts may improve durability and decrease patients' immune reactions.^{52,152,153} Recently, specifically decellularized and never-frozen homografts underlined those promising results regarding durability and patients' immune reactions.^{152,153}

With the renewed interest in the Ross procedure in the current era, a worldwide shortage of pulmonary homografts can be anticipated. Efforts of transplant centres to maximize the harvesting of homografts from non-transplantable hearts should be encouraged. Furthermore, the establishment and expansion of tissue banking systems globally is essential to ensure the broader applicability of the Ross procedure. Ultimately, tissue-engineered solutions might provide the true long-lasting solution to the right-sided substitute, both in terms of supply and long-term durability.

6. REINTERVENTIONS AFTER THE ROSS PROCEDURE

Despite previous concerns of high reoperation rates after the Ross procedure,^{18,19} the current evidence demonstrates an acceptable risk of around 1%/patient-year for reintervention after the Ross procedure.^{34,58} An analysis of the GRR by Aboud et al., which included 2,444 patients, reported 84.7% freedom from any Ross-related reintervention at 15 years.¹³ In a large recent meta-analysis by Shimamura et al., including 4,910 subjects, freedom from any reintervention was 92.6% at 10 years and 80.8% at 20 years.²¹ Notenboom et al. documented a 71.1% freedom from any reinterventions at 25 years postoperatively.³⁵

6.1 Autograft failure after the Ross procedure

Autograft failure remains an important issue influencing long-term outcomes of the Ross procedure. In the following section, we elaborate on the incidence, mechanisms, predictors, and management of autograft failure following the Ross procedure.

technique, and follow-up duration. The failure typically manifests as autograft dilation, valve regurgitation, or both, and occurs in approximately 10% to 30% of patients within 10 to 25 years postoperatively. David et al. reported a cumulative incidence of autograft reoperation of 11.5% at 20 years.⁴² In the largest published multicentre report by Aboud et al., which included 2444 patients, the risk for autograft reintervention was 0.69%/patient-year.¹³ Estimated freedom from autograft reintervention at 5, 15, 20, and 25 years was 97%, 90%, 84%, and 70%, respectively. In a recent meta-analysis by Shimamura et al., aggregated freedom from autograft reintervention at 5, 10, 15, and 20 years was 98%, 95%, 91%, and 85%, respectively.²¹

6.1.2 Risk factors for autograft failure. In several large series with long-term follow-up, preoperative AR and annulus dilatation arise as the main risk factors for autograft failure independent of the surgical technique used.^{13,41,42,99} Underlying genetic tissue abnormalities present in these conditions were thought to predispose the PA to dilate under systemic pressures over time. Surprisingly, for many of those studies, various surgical techniques and annulus reinforcement methods were not able to attenuate the negative impact of these risk factors. However, annuloplasty techniques were heterogeneous and not applied systematically. Yet, other studies analysing the effect of autograft reinforcement (at both the annulus and the STJ levels) performed more systematically were able to mitigate the impact of AR and a dilated aortic annulus on long-term outcomes.^{22,89,92,122,134} Whether or not the up-to-date surgical techniques (i.e., tailored free-standing root, native inclusion, and prosthetic inclusion) will be able to neutralize the known risk factors for autograft failure in the long run warrants further investigation.

6.1.3 Modes of autograft failure. Autograft failure occurs due to the development of autograft regurgitation (structural valve degeneration), dilatation of the PA (non-structural valve degeneration), or a combination of both.^{42,132,154} Early autograft

failure, featuring insufficiency only, is caused mainly by technical failure, inducing valve distortion during implantation, unrecognized PV anomalies (e.g., large fenestrations), or uncontrolled hypertension. Later failures are usually the consequences of an uncorrected mismatch between aortic and pulmonary size or insufficient autograft support. The mechanisms leading to regurgitation involve annulus or STJ dilation, whereas dilatation of the sinus segment of the autograft can occur without developing insufficiency. Autograft dilatation is linked to exposure to higher systemic pressures that it was not originally adapted to handle. Although tissue remodelling seems to lead to desirable valvular adaptation, adaptation of the PA root wall may be less effective.¹⁵⁵ In certain patients, especially those with AR, a large aortic annulus, and proximal aorta, the unsupported PA may dilate early in the postoperative period with subsequent valvular dysfunction.^{156,157} Localized dissection of dilated PA is rare but has been described in a few case reports.^{158–160}

6.1.4 Management of autograft failure. Whether the autograft fails by means of valve insufficiency or autograft dilatation, the valve cusps remain thin and pliable in many cases. Calcific degeneration of the autograft valve leading to stenosis is extremely rare. Autograft valve sparing procedures (aortic root remodelling with annuloplasty or autograft valve reimplantation) may be performed in patients with autograft dilation and good cusp quality (i.e., thin and pliable cusps without restriction). Less frequently, repair of the autograft cusps in patients with isolated cusp prolapse may be considered with optional supracoronary aortic replacement and/or aortic annuloplasty. In all other instances, mAVR or bAVR is performed according to patient preference and characteristics. The reoperations after the Ross procedure may be complex, and a high degree of surgical expertise is necessary to achieve acceptable outcomes.

In patients with severe PA regurgitation, the indication for surgery follows the same guidelines as for native AR. Hospital mortality after any Ross-related reoperation varies from 0% to 15% in the literature, with a mortality rate close to 0% to 1% after the first autograft reoperation, particularly in high-volume centres.^{12,122,161–165} The rate of valve-sparing procedures for autograft failure varies from 20% to 80%, depending on the type of failure and the surgeon's expertise with valve-sparing procedures.^{122,161,162,166} In a large single-centre study, Jahanyar et al. documented a series of 63 patients having autograft valve-sparing procedures.¹⁶⁴ Indications for an autograft reoperation were dilatation in 27%, dilatation plus insufficiency in 43%, and isolated insufficiency in 30%. A total of 76% of the patients had a free-standing RR as the initial Ross procedure. After a median follow-up of 8 years, the cumulative survival was 92% at 10 years, and freedom from a second autograft reoperation was 80%. In another multicentre study, Mookhoek et al. reported on 86 patients from 6 European centres.¹⁶³ In these patients, a valve-sparing

reoperation was performed a median of 9 years after the Ross procedure. The cumulative survival was 89% at 8 years, and freedom from a second autograft reoperation was 76%. Patients with severe autograft regurgitation had an increased hazard of reoperation after a valve-sparing reoperation. For these patients, careful preoperative weighing of surgical options is required.^{162,163}

6.2 Pulmonary homograft and xenograft failure after the Ross procedure

The risk of failure of the right-sided conduit adds to the overall risk of reoperation after the Ross procedure. In paediatric Ross patients, the risk of right-sided conduit reinterventions is higher compared to that of adult patients because of conduit size, child somatic growth, a more intense immune reaction in infants, and the frequent use of bovine jugular conduits, which degenerate faster compared to the substitutes used in adults (homograft or xenograft).¹⁵³ Freedom from right-sided reinterventions in adult Ross patients is 96% at 10 years, 90% at 15 years, and 88% at 20 years, which is higher compared to autograft reinterventions in the same population.^{143,167} Importantly, although the risk of reoperation may be limited and the results of reoperations are excellent, homograft dysfunction without the need for surgery or intervention may be more common.⁵⁸

The risk of homograft endocarditis is relatively low, with only 0.2%/patient year.¹⁴³ Early homograft degeneration is observed in the first 18 to 24 months after the Ross procedure, which is possibly due to technical errors (torsion, traction) but is more likely a result of immune processes.^{144,151,152} These changes are followed by a low and constant risk of homograft degeneration in the long term, featuring progressive calcification, stenosis, and regurgitation. Patients who develop an RVOT peak gradient >40 mmHg and right ventricle dilatation/dysfunction or symptoms should be considered for reintervention. The treatment options include reoperation with a new homograft or a transcatheter intervention with either a self-expandable valve or a balloon-expandable system.^{168–170}

A right-sided conduit reoperation requires careful planning, including chest CT angiography. When removing the calcified degenerated homograft, care must be taken not to injure the juxtaposed autograft and LAD posteriorly. To avoid injury, the posterior homograft wall can be left in place.

If transcatheter intervention is planned on the RVOT, specific measurements should assure that the novel transcatheter heart valve has enough space to expand, which may be hindered by severe circular calcifications. RVOT rupture may occur if balloon dilatation is too aggressive.¹⁷¹ In addition, care must be applied to avoid injuring the left main stem by balloon inflation. Prestenting of the RVOT might be important, and a low post-interventional transvalvular gradient is a predictor of a satisfactory and durable long-term outcome.¹⁷²

Consensus Statement Table 4. Reinterventions After the Ross procedure
• Valve-sparing procedures to treat a failing autograft may be considered as an alternative to valve replacement. • Reintervention on a failing pulmonary homograft/xenograft may often be performed percutaneously.

7. POSTOPERATIVE MANAGEMENT

To maximize the long-term benefits of the Ross procedure, lifelong postoperative surveillance is mandatory. The specifics of follow-up after the Ross procedure comprise blood pressure management, anti-inflammatory drugs, and annual surveillance echocardiograms.

7.1 Blood pressure control

Strict blood pressure control is a fundamental step in the prevention of autograft failure in the short- and long-term postoperatively. After transfer to the aortic position, the PA is immediately exposed to systemic blood pressure. This condition initiates a process of adaptive remodelling in the autograft valve and wall. Endothelial and interstitial cells undergo activation and phenotypic changes, allowing the autograft to mimic the complex anatomy and function of the native aortic root.^{27,173} This process takes place in the first months and years postoperatively, and, until that time, the autograft is vulnerable to dilation and irreversible damage.¹⁵⁶ In the absence of prospective randomized evidence, the authors suggest that keeping the systolic blood pressure below 110–115 mmHg is desirable to minimize the risk of autograft dilation in the first year postoperatively.^{17,23}

Indeed, it has been demonstrated that early autograft dilatation observed at the time of hospital discharge is associated with late autograft dilation in some patients.¹⁵⁶ In other words, early dilatation begets late dilatation. Therefore, blood pressure control must begin immediately after cross-clamp release and continue very tightly thereafter. Bouhout et al. reported that a tailored surgical approach in combination with a strict blood pressure control protocol effectively stabilizes the autograft dimensions in the short- and mid-term postoperatively.⁸⁹ Particular attention must be paid to patients with pre-existing risk factors for autograft failure, such as AR and aortic annulus dilation.^{13,31} To the best of our knowledge, there is currently no evidence with regard to antihypertensive protocol in patients after native or prosthetic inclusion technique.

There is no evidence regarding the choice of antihypertensive agents specifically after the Ross procedure for the prevention of PA failure. However, some insights can be extrapolated from the guidelines on perioperative medication^{174,175} and trials including patients with thoracic aortic aneurysms and/or connective tissue disorders. Beta-blockers are suggested as a first-line treatment because they are able to reduce the dP/dT (pressure change in the aorta over time) and aortic wall stress. Studies have shown their desirable impact on the long-term prevention of aortic dilation.^{176,177} If another drug is necessary (due to intolerance of beta-blockers or insufficient hypertension control), angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers have proven to be valuable alternatives.^{23,89,178–180} After the first postoperative year, reassessment of medication in line with general guidelines for management of hypertension is desirable.¹⁸¹

Expanding the standards of blood pressure measurement beyond the outpatient clinic, with the utilization of home self-measurement, mobile apps, or dedicated nurses for regular check-ups, can have an additional effect on patient compliance and effective blood pressure control.^{89,181}

7.2 Anti-inflammatory drugs

The inflammatory response has been considered an important contributor to pulmonary homograft failure. The host reaction against the homograft is most pronounced in the early postoperative phase and may involve the proximal and distal suture line and cusps, ultimately resulting in pulmonary stenosis.^{144,182} The risk of homograft failure follows a biphasic pattern, reaching 3.5% in the first year postoperatively and then falling to 1% per year in the long term.⁵²

To reduce the potential detrimental effect of nonspecific inflammation on pulmonary homograft in the early postoperative period, several groups have proposed administration of nonsteroidal anti-inflammatory drugs in the first 3 to 6 months postoperatively.^{15,52,183} Limited evidence has shown that ibuprofen intake (5 mg/kg 3 times daily) may significantly improve the longevity of a pulmonary homograft.¹⁸³

There is no evidence to support the administration of aspirin after the Ross procedure. A single study recommended switching to lifelong aspirin after the initial ibuprofen period.¹⁸⁴ Nevertheless, general practice is to withdraw any anti-inflammatory drugs after the early postoperative period and leave the patient without chronic medication.

7.3 Long-term surveillance

Patients after the Ross procedure require lifelong surveillance, ideally at a tertiary care centre with a dedication to aortic root surgery. Apart from regular clinical evaluation, echocardiographic follow-up represents a crucial part of the long-term postoperative patient management. Early detection of autograft/homograft dysfunction, followed by timely intervention, if necessary, is very important to sustain the long-term patient benefit and minimize the risk of adverse events.

We propose the first postoperative echocardiographic examination at hospital discharge, then at 3 to 6 months postoperatively, and once a year thereafter. Echocardiographic parameters of interest after the Ross procedure include the diameters of the aortic (i.e., autograft) annulus, aortic root, STJ, and ascending aorta; the degree and characteristics of AR; gradient over autograft; the degree and characteristics of pulmonary regurgitation; gradient over homograft; left ventricular ejection fraction; left ventricular end-systolic and end-diastolic diameter; and right ventricular ejection fraction. In unclear cases, speckle tracking analysis or cardiac MR may provide additional information about left and right ventricular volume and function.^{20,52,185} TOE and cardiac CT may also play a role in the advanced morphologic examination of the complete scale of pathologies, particularly in ruling out endocarditis.^{57,186}

Consensus Statement Table 5. Postoperative Management

- Meticulous blood pressure control is desirable in the first year after the Ross procedure to reduce the risk of autograft dilation.
- Administration of oral anti-inflammatory medications in the first 3–6 months after the Ross procedure may reduce the risk of pulmonary homograft failure.
- Lifelong clinical and echocardiographic follow-up is mandatory after the Ross procedure.

8. ROSS CENTRES OF EXCELLENCE AND HOW TO START

As the evidence demonstrating excellent outcomes after the Ross procedure grows, so does the enthusiasm for the Ross procedure globally in the surgical community. The recognition that the Ross procedure may restore patient life expectancy to that of the general population has also led to an increasing demand among patients and cardiologists. Access to the Ross procedure, however, remains limited, resulting in significant geographic and socioeconomic barriers for patients. Lack of easy access to the Ross procedure has been cited as a reason why international guideline documents have not more strongly recommended the Ross procedure despite the overwhelming evidence supporting it.

The Ross procedure is undoubtedly a technically more complex operation than conventional AVR. A volume–outcome relationship has been observed for many cardiac surgery procedures, including mitral valve repair and aortic root replacement.^{187,188} When performed in low-volume centres, the perioperative mortality of the Ross procedure is 3-fold higher than that of AVR.¹⁹ There is, however, no evidence of a higher mortality risk when the Ross procedure is performed in high-volume centres.^{13,20,21} Recent data from the Society of Thoracic Surgeons Registry has shown a steep incline in the utilization of the Ross procedure since 2017.

multimodality imaging, dedicated perioperative TOE and cardiac anaesthesia, a cardiologist with valve expertise, and an expert in perioperative management of these patients. Ross Centres of Excellence should meet prespecified metrics of case volumes, operative mortality, early clinical and echocardiographic outcomes, and longitudinal follow-up. The establishment of these centres requires a dedicated team to develop and maintain expertise in a focused, concentrated manner. A learning curve has been observed for the Ross procedure, whereby after approximately 75 cases, the risk of morbidity is minimized.^{190,191} Initiation of a new Ross program should be done with the appropriate training and peer-to-peer support from expert surgeons. Ross courses with various formats (didactic, wet labs, live cases), training programs (observerships or fellowships), and proctoring for the initiation of a program are potential avenues of training for surgeons interested in establishing a Ross program. Congenital surgeons are also a source of expertise for adult Ross programs that are often initiated by surgeons with established expertise in aortic root surgery. A well-prepared and committed surgical team is necessary to minimize the impact of the learning curve on establishing a new program. Ultimately, the viability of the Ross procedure depends on the safe expansion of the procedure to an increasing number of centres worldwide. Transparent data reporting will be the cornerstone of Ross Centres of Excellence and will permit a careful dissemination of this procedure.

Consensus Statement Table 6. Centralization and Ross Centres of Excellence

- There is a strong positive volume–outcome association for the Ross procedure at the surgeon and centre levels.
- The Ross procedure should be offered at Ross Centres of Excellence defined by case volumes, operative mortality, early clinical and echocardiographic outcomes, and longitudinal follow-up.

Nevertheless, the top 10 highest volume centres in North America accounted for 53% of all Ross procedures performed between 2008 and 2023.¹⁸⁹ Furthermore, a volume–outcome relationship for the Ross procedure was observed at both the centre-level and the surgeon-level. In-hospital mortality was significantly higher at centres performing 1 or 2 Ross procedures annually compared to those performing more than 10 (OR 4.5 [95% CI: 1.5–13.2], $P=.006$). Similarly, higher in-hospital mortality was observed for surgeons performing 1 or 2 Ross procedures per year compared to those performing more than 10 (OR 3.8 [1.2–12.2], $P=.007$). The technical nuances of this procedure are such that surgical expertise is a crucial determinant of both the early and late outcomes. Furthermore, patient safety is paramount and should not be compromised as enthusiasm for the Ross procedure grows.

The preceding considerations have led us to suggest that the Ross procedure be offered in Ross Centres of Excellence, which should be defined and established by medical providers or by EACTS or national surgical societies. New programs require a surgeon committed to acquiring and maintaining the expertise necessary to sustain a Ross program. Expertise in aortic root surgery and aortic valve repair is a prerequisite and goes hand in hand with the Ross program. A minimum of 10 to 15 cases per year, and ideally 20 to 25 Ross procedures per year, are required to establish a high-quality program, with the caveat that burgeoning programs may take a few years to achieve that volume. In addition to surgical expertise, these centres benefit from the infrastructure of a comprehensive valve centre, with

9. FUTURE DIRECTIONS AND GAPS IN KNOWLEDGE

Future efforts of Ross surgeons and researchers worldwide should concentrate on raising awareness about the outcomes of the Ross procedure in non-elderly patients with AV disease. This document aims to provide a comprehensive and critical review of current knowledge about the Ross procedure that will eventually encourage a move toward its wider implementation in clinical practice guidelines.

A transparent, detailed, and longitudinal collection of robust data on the outcomes of the Ross procedure is the cornerstone of clinical research in this field. It represents a firm basis for future clinical recommendations. Large multicentre registries and meta-analyses of current data will provide further insight into the real-world results of patients undergoing the Ross procedure. Analysis of potential risk factors for reinterventions and other major events is of particular interest, because their detailed scientific evaluation may lead to an improvement in patient selection, surgical techniques, and postoperative management to minimize the risks and maximize the benefits for the patients.

Future research should concentrate particularly on the long-term outcomes of contemporary surgical techniques of PA implants (tailored RR, autologous and prosthetic inclusion, Ross-PEARS) and their ability to prevent autograft reinterventions in the high-risk patient population (with dilated aortic annulus and/or AR). An important field of research should aim at

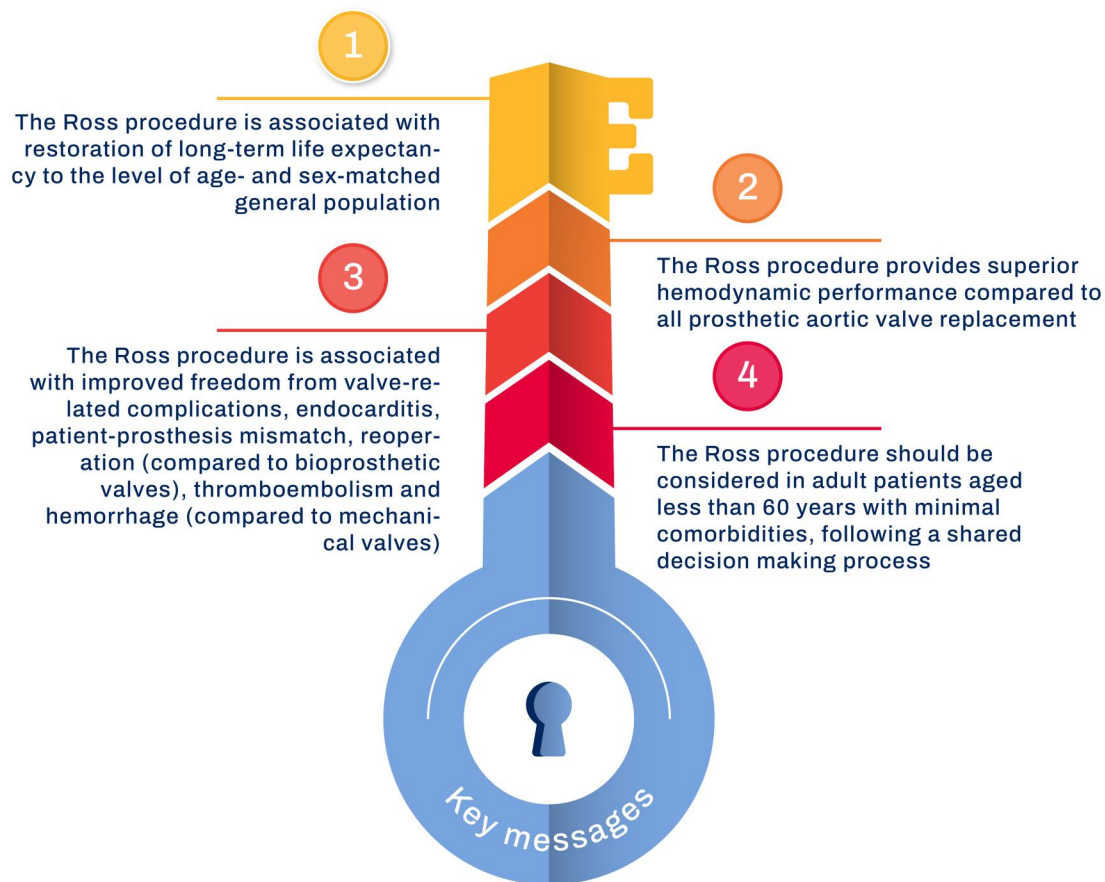


Figure 11. Key Messages from the European Association of Cardio-Thoracic Surgery Expert Consensus Document on the Ross Procedure

improving the long-term outcomes of right-sided conduits, further development and widespread use of decellularized pulmonary homografts, and/or the development of novel PV substitutes, ideally at off-the-shelf disposal. The evidence to support postoperative anti-inflammatory and antihypertensive treatment is minimal; future research should examine the importance of postoperative management in a more robust and systematic way.

Ultimately, there is a constantly ongoing debate about the lack of prospective randomized trials on the best AV substitute in the non-elderly patient population. The only RCT with long-term follow-up by El-Hamamsy et al. compared the Ross procedure with aortic homograft, which does not reflect current valve replacement options and contemporary practice.¹⁵ A recent attempt to prospectively compare the Ross procedure with conventional AVR options, the REVIVAL trial, was abandoned prematurely due to challenges with randomization in this patient population and lack of equipoise.¹⁹² Although an RCT would eliminate the issue of patient selection and inherent bias associated with retrospective studies, significant concerns remain with regard to the willingness of patients, referring cardiologists, and surgeons to participate in such research in light of current evidence (as was recently the case of an RCT comparing valve-sparing root replacement versus mechanical composite valve graft replacement.¹⁹³). Analogically, there is no RCT to compare mitral valve reconstruction and replacement in patients with primary degenerative mitral valve disease. However, valve reconstruction is generally considered a preferred option based on robust retrospective data.

10. CONCLUSIONS

This expert consensus document summarizes and critically reviews current knowledge on the use of the Ross procedure in adult patients and provides clinical statements to assist clinicians and patients in treatment decision-making and daily practice (Figure 11). The Ross procedure is the only living valve substitute for patients with unreparable AV disease. In experienced hands, it is associated with the restoration of long-term survival up to that of the general population. It has low thrombogenicity, excellent haemodynamics, and good resistance to infection. The risk of reintervention is acceptable if an appropriate surgical technique is used. Patient selection and meticulous preoperative examination are crucial to provide the desired patient benefit.

Tailored root replacement and native and prosthetic root inclusion are currently the most widespread techniques. Postoperative management based on regular surveillance, blood pressure control, and transient nonsteroidal anti-inflammatory drug administration is an integral part of patient care. The establishment and the expansion of tissue banking systems globally are essential to ensure broader accessibility of the pulmonary homografts and thus the Ross procedure. There is a strong volume–outcome association for the Ross procedure at the surgeon- and centre-levels. The Ross procedure should be concentrated in Ross Centres of Excellence defined by case volumes, operative mortality, early clinical and echocardiographic outcomes, and longitudinal follow-up. Future research should focus on the analysis of very long-term postoperative data, the outcomes and improvement of

contemporary PA implant techniques, and further development of pulmonary root replacement substitutes. More robust and systematic evidence evaluating all aspects of the procedure, including risk factors, preoperative planning, surgical techniques, and post-operative management, will be necessary to further improve the outcomes of the Ross procedure in general.

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DATA AVAILABILITY STATEMENT

No new data were generated or analysed in support of this research.

CONFLICTS OF INTEREST STATEMENT

Martin Andreas reports receiving consulting fees from Edwards, Abbott, Medtronic, AbbVie, LSI, and B. Braun. Joseph E. Bavaria reports receiving consulting fees from Artivion. Filip Casselman reports receiving payment from Edwards Lifesciences for lectures. Filip Casselman reports a leadership role as Chair of the Acquired Cardiac Disease Domain at EACTS. Laurent de Kerchove reports receiving consulting fees from Corcym for a preceptorship (paid to the institution). Ismail El-Hamamsy reports participation on an advisory board at the Aqueadon Duett Vascular Study DMSB. Ismail El-Hamamsy reports leadership positions as president of the Heart Valve Society and member of the Board of Directors at the American Association for Thoracic Surgery. Tomas Holubec reports receiving consulting, speaker honoraria and travel compensations from Abbott, Artivion, Edwards Lifesciences, Getinge, Sweden and SmartCanula LLC, Switzerland. Milan Milojevic reports a leadership position as Director of the Clinical Practice Guidelines Programme at the European Association for Cardio-Thoracic Surgery. Maral Ouzounian reports leadership positions as President of the Canadian Society of Cardiac Surgeons and as Secretary-Elect of the American Association for Thoracic Surgery (unpaid). Peter Verbrugge reports receiving consulting fees and speaker honoraria from Edwards Lifesciences, Medtronic, Artivion and Corcym. All other authors have declared that they have no conflicts of interest.

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11. SUPPLEMENTARY MATERIAL

[Supplementary data](#) are available at the *European Journal of Cardio-Thoracic Surgery* online.

12. REFERENCES

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